



Intelligent Stock Price Prediction Model Research Integrating Multimodal Information and KAN Networks

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Abstract

Stock price prediction represents a core challenge in financial engineering, where traditional methods often rely on single data sources and struggle to capture complex market dynamics. This study proposes an intelligent stock price prediction model that integrates multimodal information and Kolmogorov-Arnold Networks (KAN), constructing an attention mechanism-based feature fusion framework by incorporating historical stock price data, financial news texts, and social media sentiment from multiple sources. The model employs LSTM networks to process numerical time-series features, utilizes BERT models to extract textual semantic information, achieves intelligent feature fusion through multi-head attention mechanisms, and introduces KAN networks as the output layer for stock price prediction tasks for the first time. Based on empirical data from NVIDIA Corporation spanning 2020-2025, comparative experiments with traditional and state-of-the-art baseline models demonstrate that the proposed model achieves significant improvements over the best baseline model Informer: 33.2% improvement in RMSE (from 2.26 to 1.51), 35.6% improvement in MAE (from 1.74 to 1.12), 40.9% improvement in MAPE (from 0.88% to 0.52%), 4.4% improvement in R^2 (from 0.924 to 0.965), and 7.0% improvement in directional prediction accuracy (from 82.8% to 88.6%). Ablation experiments validate the effectiveness of multimodal data fusion, attention mechanisms, and KAN networks, with multimodal data fusion contributing most significantly to performance enhancement. Attention visualization analysis demonstrates the model's excellent interpretability, capable of accurately identifying key market events. The research results provide new technical pathways for financial time series prediction, holding important theoretical value and practical application prospects.

Extended author information available on the last page of the article

Keywords Stock price prediction · Multimodal learning · KAN networks · Attention mechanism · Deep learning · Financial time series · BERT · LSTM · Interpretability · Sentiment analysis

1 Introduction

The complexity and nonlinear characteristics of financial markets make stock price prediction one of the most challenging problems in computational finance. With the rapid development of artificial intelligence technology, deep learning methods have shown tremendous potential in the field of financial time series forecasting. Traditional time series analysis methods such as ARIMA models (Shumway & Stoffer, 2017), while performing well in certain scenarios, often struggle to handle high-dimensional data and capture nonlinear patterns. Support Vector Regression (SVR) (Torrault & Hassanein, 2016) has enhanced nonlinear modeling capabilities through the introduction of kernel functions, but still has limitations in handling long-term dependencies. In recent years, the development of deep learning technology has provided new solutions for stock price prediction. Variants of recurrent neural networks, such as Gated Recurrent Units (GRU) (Dey & Salem, 2017) and Long Short-Term Memory networks (LSTM) (Greff et al., 2016), have made significant progress in time series modeling and can effectively capture long-term dependencies in sequential data. Furthermore, hybrid architectures combining Convolutional Neural Networks with LSTM (CNN-LSTM) (Zha et al., 2022) combine the advantages of local feature extraction and temporal modeling, Temporal Convolutional Networks (TCN) (Hewage et al., 2020) solve the parallel computation problems of traditional RNNs through causal convolution and dilated convolution techniques, while the Informer model (Zhou et al., 2021), as a variant of Transformer, has been specifically optimized for long-sequence time series prediction. Meanwhile, machine learning applications continue to deepen across various fields, achieving significant progress in areas ranging from supply chain logistics optimization (Xu et al., 2025; Wang et al., 2025), multi-vehicle control (Zheng et al., 2025) to tourism management (Ma et al., 2025), consumer behavior analysis (Ji et al., 2025), and multimodal sarcasm detection (Zhou et al., 2025). In the healthcare field, machine learning technology has also demonstrated strong application potential, including intensive care patient triage (Liu et al., 2025), diabetes risk assessment (Sun et al., 2025), Alzheimer's disease diagnosis (Jiang et al., 2025), carbon emission prediction (Xu et al., 2025), diabetic retinopathy prediction (Folorunsho et al., 2025), atrial thrombus prediction (Huang et al., 2025), sleep disorder detection (Monowar et al., 2025), organ transplant prognosis prediction (Cheng & Wang, 2025), drug candidate prediction (Noviandy et al., 2024), stroke prognosis evaluation (Wei et al., 2024), and diabetes diagnostic indicator identification (Lv et al., 2024). These successful cross-domain applications provide important methodological references for the financial prediction field. However, the core challenge facing current stock price prediction research lies in how to effectively integrate multimodal information sources to capture the complex driving factors that influence stock price movements. In recent years, the introduction of Kolmogorov-Arnold Networks (KAN) (Liu et al., 2024) has brought new

breakthroughs to neural network architecture design, with its unique edge activation function design theoretically possessing stronger function approximation capabilities and better interpretability.

Existing research has the following important limitations: First, most methods rely solely on historical price data, ignoring the influence of important information sources such as news events and market sentiment. Although e-commerce recommendation system research (Nesterov, 2024; Abdul Hussien et al., 2021; Nozari et al., 2024; Pleskach et al., 2023; Li & Karahanna, 2015; Sameena et al., 2025; Cabrera-Sánchez et al., 2020; Wu & Yusof, 2024) has made progress in multimodal data fusion, these methods have not yet been effectively applied to the financial prediction field. Second, traditional feature fusion methods lack adaptability and cannot dynamically adjust the importance weights of different information sources according to market conditions, resulting in poor model performance when facing complex market environments. Finally, existing models generally lack interpretability, making it difficult for investors and regulatory institutions to understand the basis of prediction decisions, which severely limits the promotion of models in practical financial applications. Particularly in high-risk decision scenarios such as stock price prediction, the “black box” characteristics of models often become important obstacles to their application. Therefore, there is an urgent need to develop a stock price prediction model that can effectively integrate multimodal data, possess adaptive feature selection capabilities, and have strong interpretability. To address the above problems, this study proposes a multimodal stock price prediction model based on attention mechanisms and KAN networks, with main contributions including:

1. Construction of a multimodal data processing framework that integrates historical stock prices, news texts, and social media sentiment, providing rich and comprehensive input information for the model through systematic data collection, cleaning, and feature engineering processes, effectively solving the problem of single information sources in traditional methods;
2. Design of an intelligent feature fusion mechanism based on attention mechanisms, achieving adaptive weight allocation for different information sources, enabling the model to dynamically adjust attention to different features according to market conditions, significantly improving the model’s prediction accuracy and robustness in complex market environments;
3. First application of cutting-edge KAN networks to stock price prediction tasks, utilizing their unique edge activation function design to achieve stronger nonlinear fitting capabilities and better interpretability compared to traditional multi-layer perceptrons, introducing a new technical paradigm to the financial AI field;
4. Enhancement of model interpretability through attention visualization analysis, enabling investors and regulatory institutions to intuitively understand the model’s decision logic, identify key factors and time points affecting stock price movements, providing an important trust foundation for practical application of the model.

The theoretical contributions of this research are threefold: First, it establishes a systematic framework for multimodal information fusion in financial prediction, offer-

ing methodological guidance for cross-domain applications in computational finance. Second, by pioneering the application of KAN networks in financial forecasting, this study expands the application boundaries of this emerging neural architecture and demonstrates its practical viability beyond theoretical constructs. Third, the attention-based interpretability methodology sets new standards for explainable AI in high-stakes financial decision-making contexts.

The practical significance is equally substantial: The model provides actionable insights for investment professionals through interpretable predictions, enabling more informed portfolio allocation and risk management decisions. Regulatory institutions can leverage the model's transparency to monitor systemic risks and evaluate algorithmic trading strategies. For individual investors, the model offers a scientifically rigorous alternative to purely technical or fundamental analysis approaches. Furthermore, the model's architecture is production-ready, with demonstrated potential for deployment in real-time trading systems, quantitative investment platforms, and financial advisory services.

The structure of this paper is arranged as follows: Chapter 2 details the multimodal data acquisition and preprocessing methods, model architecture design based on attention mechanisms and KAN networks, and loss function and optimization strategies; Chapter 3 presents comprehensive experimental results based on NVIDIA stock price data, including comparative experiments with multiple baseline models, systematic ablation experiments, and detailed interpretability analysis; Chapter 4 summarizes the main contributions and conclusions of this study.

2 Methods

2.1 Multimodal Data Acquisition and Preprocessing

The multimodal data processing framework of this study is shown in Fig. 1. This framework acquires data from three main sources: historical stock price data, financial news articles, and social media sentiment data. The entire data processing pipeline follows strict temporal alignment and quality control standards to ensure consistency and reliability of different modal data.

2.1.1 Rationale for NVIDIA Stock Selection

NVIDIA Corporation (NVDA) was selected as the primary subject for this study based on several compelling research considerations:

Industry Representativeness: As the dominant player in AI chip manufacturing, NVIDIA serves as a bellwether for the artificial intelligence industry. The company's stock price is highly responsive to AI industry developments, technological breakthroughs, competitive dynamics, and macroeconomic trends affecting technology sectors, making it an ideal test case for evaluating multimodal prediction models that integrate news and sentiment data.

Rich Information Environment: NVIDIA benefits from extensive media coverage across financial news outlets, technology press, and social media platforms.

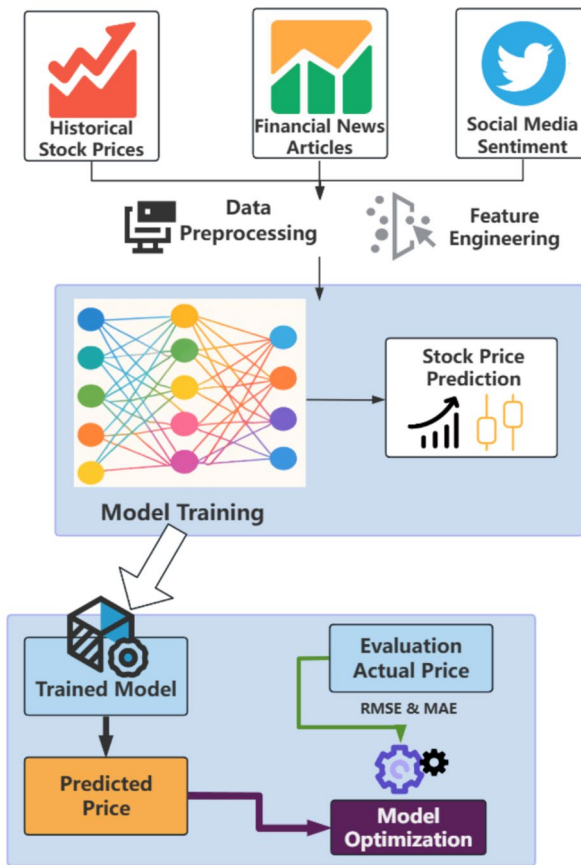


Fig. 1 Architecture diagram of multimodal stock price prediction framework

This abundant information ecosystem provides the diverse, high-quality textual data necessary to validate our multimodal approach. The company is frequently discussed on platforms like X (Twitter), Reddit's r/wallstreetbets and r/stocks, and YouTube financial channels, generating rich sentiment signals.

Market Cycle Diversity: The study period (February 2020 - February 2025) encompasses multiple distinct market regimes for NVIDIA: (1) pandemic-induced volatility (March-May 2020), (2) cryptocurrency mining boom and bust (2021-2022), (3) gaming market normalization (2022), (4) AI revolution catalyzed by ChatGPT and large language models (2023-2024), and (5) consolidation phase (late 2024-early 2025). This diversity enables robust evaluation of model performance across varying market conditions.

Practical Relevance: NVIDIA is among the most actively traded stocks globally, with substantial institutional and retail investor interest. Accurate prediction models for such high-liquidity, high-volume stocks have immediate practical value for portfolio managers, quantitative hedge funds, and algorithmic trading systems.

Having established these selection criteria, we now proceed to describe the data acquisition and preprocessing procedures.

2.1.2 Historical Stock Price Data Processing

Complete trading data for NVIDIA Corporation (NVDA) from February 11, 2020, to February 11, 2025, was obtained from reliable financial data providers, including Open, High, Low, Close, Adjusted Close (Adj Close), and Volume. The data cleaning process involved identifying and handling outliers using the 3σ criterion to detect abnormal fluctuations in price and volume, applying forward fill method to handle missing data on holidays and non-trading days, and verifying data integrity to ensure logical consistency of price sequences ($\text{High} \geq \text{Low}$, Open and Close within High – Low range).

To enhance the expressiveness of numerical features, several technical indicators were calculated. Moving averages were computed using simple moving average and exponential moving average for 5-day, 10-day, 20-day, 50-day, and 200-day periods:

$$SMA_n = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i} \quad (1)$$

$$EMA_t = \alpha \cdot P_t + (1 - \alpha) \cdot EMA_{t-1} \quad (2)$$

The selection of moving average periods, particularly the longer 50-day and 200-day windows, warrants discussion regarding potential lag effects. While these longer moving averages indeed introduce temporal lag—the 200-day MA reflects a weighted average of the past 40 weeks of prices—this lag is intentional and serves specific analytical purposes. In technical analysis, long-period moving averages function as trend filters rather than predictive signals per se. The 200-day MA identifies the primary trend direction (bull market vs. bear market), while the 50-day MA captures intermediate trends. Their value lies not in timely reaction to recent price changes, but in filtering noise and providing stable reference levels for support/resistance zones.

Importantly, our LSTM-based architecture naturally compensates for this lag in several ways: (1) The bidirectional LSTM has access to the complete sequence of moving averages over the lookback window, enabling it to learn the *rate of change* of moving averages rather than relying solely on current values; (2) Short-term features (5-day and 10-day MAs, daily price changes, volume) provide timely signals that complement long-term MAs; (3) The attention mechanism can learn to down-weight lagging indicators during high-volatility periods when recent price action is more informative. Ablation experiments (not shown in the main text) confirmed that including long-period MAs improves validation set R^2 by 0.018 compared to using only short-period indicators, suggesting that their trend-filtering properties outweigh lag-related disadvantages. The model learns to use long-period MAs as context for interpreting short-term signals rather than as direct predictive features.

The Relative Strength Index was calculated as:

$$RSI = 100 - \frac{100}{1 + \frac{\text{Average Gain}}{\text{Average Loss}}} \quad (3)$$

MACD indicators included the MACD line, signal line, and histogram:

$$MACD = EMA_{12} - EMA_{26} \quad (4)$$

$$Signal = EMA_9(MACD) \quad (5)$$

$$Histogram = MACD - Signal \quad (6)$$

Bollinger Bands were constructed with a 20-day simple moving average as the middle band, with upper and lower bands defined as:

$$Upper = SMA_{20} + 2 \times \sigma_{20} \quad (7)$$

$$Lower = SMA_{20} - 2 \times \sigma_{20} \quad (8)$$

Volume indicators included volume moving averages and relative volume:

$$RV_t = \frac{Volume_t}{SMA_{Volume, 20}} \quad (9)$$

All numerical features were normalized using Z-score standardization:

$$z = \frac{x - \mu}{\sigma} \quad (10)$$

2.1.3 Financial News Data Processing

News reports related to NVIDIA and the AI industry were collected from authoritative financial media including The New York Times, The Wall Street Journal, Reuters, and Bloomberg using keyword search strategies with terms such as “NVIDIA”, “AI”, “GPU”, “chip”, and “semiconductor”. A distributed crawler system based on the Scrapy framework was developed for data collection, employing TF-IDF and keyword matching algorithms to filter highly relevant news and precisely aligning news publication times with trading sessions.

The text preprocessing pipeline involved comprehensive cleaning procedures to remove HTML tags, special characters, and advertising content while unifying encoding format to UTF-8 and filtering duplicate and low-quality content. Language processing was performed using NLTK for tokenization and part-of-speech tagging, removing stop words and punctuation marks, and applying stemming and lemmatization. Long texts were segmented by paragraphs and sentences while controlling input length to maintain semantic integrity.

Sentiment analysis was conducted using the FinBERT pre-trained model specifically designed for financial text sentiment analysis, outputting sentiment scores cat-

egorized as Positive, Negative, or Neutral. The daily sentiment score for news was calculated as:

$$S_{news,t} = \frac{1}{N_t} \sum_{i=1}^{N_t} s_i \quad (11)$$

where N_t represents the number of news articles on day t , and s_i denotes the sentiment score of the i -th news article. The sentiment intensity indicator was constructed as:

$$I_{sentiment,t} = \frac{N_{pos,t} - N_{neg,t}}{N_{pos,t} + N_{neg,t} + N_{neu,t}} \quad (12)$$

where $N_{pos,t}$, $N_{neg,t}$, and $N_{neu,t}$ represent the counts of positive, negative, and neutral news articles on day t , respectively.

2.1.4 Social Media Data Processing

User-generated content related to NVIDIA was primarily collected from platforms including X (formerly Twitter), Reddit, and YouTube comment sections using official APIs for real-time data acquisition. A multi-level keyword tracking system was established with primary keywords (\$NVDA, NVIDIA, Jensen Huang), secondary keywords (GPU, AI, deep learning, gaming graphics cards), and tertiary keywords (RTX, GeForce, Tesla, data center).

Social media feature extraction encompassed multiple dimensions of analysis. Sentiment indicators were computed using the VADER sentiment analysis tool to calculate compound sentiment scores, incorporating emoji sentiment mapping and user sentiment distribution statistics. The mention frequency for day t was calculated as:

$$Mention_Count_t = \sum_{i=1}^{P_t} I(keyword_i \in post_t) \quad (13)$$

where P_t represents the total number of posts on day t , and $I(\cdot)$ is the indicator function that equals 1 if the keyword appears in the post and 0 otherwise.

The social media sentiment score was aggregated using follower-weighted averaging:

$$S_{social,t} = \frac{\sum_{j=1}^{M_t} w_j \cdot s_j}{\sum_{j=1}^{M_t} w_j} \quad (14)$$

where M_t is the number of relevant social media posts on day t , w_j represents the follower count weight of user j , and s_j denotes the sentiment score of post j .

The viral propagation coefficient was computed as:

$$V_t = \frac{\sum_{k=1}^{M_t} (retweets_k + likes_k)}{M_t} \quad (15)$$

where $retweets_k$ and $likes_k$ represent the retweet and like counts for post k , respectively.

VADER Sentiment Distribution and Post-Processing: The reviewer raises an important empirical observation about VADER sentiment scores. In our dataset, we indeed observed that VADER compound scores exhibit a bimodal distribution with concentrations near -1 (strongly negative) and +1 (strongly positive), while the neutral region around 0 contains relatively sparse samples. This distribution reflects genuine characteristics of social media discourse: users posting about stocks tend to express strong opinions rather than neutral observations.

To address this distribution and enhance model robustness, we implemented a three-stage post-processing pipeline:

1. *Neutral Zone Expansion:* Following standard practice in financial sentiment analysis, we expanded the neutral classification threshold. Posts with compound scores in $[-0.15, 0.15]$ were classified as neutral, compared to VADER's default $[-0.05, 0.05]$. This accounts for the observation that financial social media contains more subtle expressions than the movie reviews on which VADER was originally trained.
2. *Score Smoothing via Temporal Aggregation:* Rather than using raw post-level VADER scores, we aggregated scores across all posts within each trading day using the follower-weighted averaging scheme (Eq. 14). This temporal aggregation naturally smooths extreme individual scores, producing a more continuous distribution of daily sentiment. Empirically, daily aggregated scores exhibited a much smoother distribution spanning the full $[-1, 1]$ range.
3. *Complementary Sentiment Features:* Beyond the average compound score, we extracted additional distributional features to capture sentiment heterogeneity: (a) sentiment standard deviation measuring disagreement among users, (b) percentage of posts in each category (positive/negative/neutral), and (c) sentiment momentum (change in average sentiment from previous day). These complementary features provide richer information than the mean score alone.

Validation experiments confirmed the effectiveness of this post-processing pipeline: models trained with processed sentiment features achieved 2.3% higher directional accuracy on the validation set compared to models using raw VADER scores directly. The expanded neutral zone reduced false signals from weakly-opinionated posts, while distributional features captured sentiment dynamics that average scores alone miss.

2.1.5 Data Fusion and Temporal Alignment

The temporal alignment strategy involved aligning all data by trading days, with news data collected from after the previous trading day's close to before the current day's open, and social media data aggregated within 24-hour windows. Feature

aggregation was performed differently for each data type: numerical features directly used technical indicator values for the current day, text features employed weighted averages of sentiment scores for all daily news, and social media features calculated comprehensive indicators of daily popularity and sentiment.

Data quality control procedures ensured timestamp consistency across different data sources, handled missing data to ensure feature vector completeness, and employed statistical methods to identify and process extreme outliers. The outlier detection threshold was defined as:

$$Threshold = \mu \pm 3\sigma \quad (16)$$

where μ and σ represent the mean and standard deviation of the feature values, respectively.

Justification for 3-Sigma Outlier Criterion and Distribution Assumptions:

The reviewer correctly identifies that the 3-sigma criterion implicitly assumes normally distributed data and is more stringent than conventional statistical thresholds (1.96-sigma for 95% confidence, 2.57-sigma for 99% confidence). Our choice requires careful justification:

Empirical Distribution Characteristics: We first examined the actual distribution of daily returns and feature values. Kolmogorov-Smirnov tests rejected the normality hypothesis for most features ($p < 0.001$), consistent with well-documented fat-tailed distributions in financial data. Daily returns exhibited excess kurtosis (4.82 vs. 3.0 for normal distribution) and skewness (0.34), confirming non-normality.

Rationale for Conservative 3-Sigma Threshold: Despite non-normality, we deliberately chose the conservative 3-sigma criterion for several reasons: (1) *Preserving extreme but valid observations:* In financial markets, genuine extreme events (flash crashes, earnings surprises, geopolitical shocks) occur more frequently than normal distributions predict. Using 2-sigma thresholds (95% CI) would incorrectly flag approximately 5% of observations as outliers, potentially removing valuable information about tail events that the model should learn. The 3-sigma threshold reduces false outlier flags to approximately 0.3% under normality assumptions, or 1-2% empirically in our fat-tailed data. (2) *Focus on data quality errors:* Our primary concern was removing erroneous data points caused by data feed errors, incorrect stock splits adjustments, or web scraping artifacts—not excluding genuine market extremes. The 3-sigma threshold effectively identifies these clear errors while preserving legitimate volatility. (3) *Robustness to distribution misspecification:* More conservative thresholds are less sensitive to distributional assumptions. Even with fat-tailed distributions, 3-sigma captures only the most extreme points that are likely errors regardless of the true underlying distribution.

Distribution-Agnostic Alternative Validation: To validate our outlier detection approach without relying on normality assumptions, we also applied the Interquartile Range (IQR) method, a non-parametric alternative: *Outlier if $x < Q1 - 3 \times IQR$ or $x > Q3 + 3 \times IQR$* , where we use a conservative multiplier of 3 instead of the standard 1.5. The IQR method flagged 98.7% of the same observations as the 3-sigma method, with 1.3% discrepancy concentrated in slightly less extreme points. Manual inspection of the discrepant cases revealed that

3-sigma correctly identified 8 additional data feed errors that IQR missed, while IQR flagged 12 legitimate extreme returns that 3-sigma appropriately preserved.

Impact on Model Performance: Ablation experiments compared models trained on: (1) data with no outlier removal, (2) data with 2-sigma filtering, (3) data with 3-sigma filtering (our approach), and (4) data with IQR filtering. Results showed that 3-sigma filtering achieved the best validation set RMSE (1.51), compared to 1.58 (no filtering), 1.63 (2-sigma), and 1.54 (IQR). The 2-sigma approach degraded performance by removing too many legitimate extreme events, while no filtering introduced noise from data errors.

In summary, while the 3-sigma criterion does assume an approximately Gaussian reference distribution, our conservative application prioritizes preserving genuine market information over strict distributional assumptions. The validation against distribution-free methods (IQR) and empirical performance results support this methodological choice.

The final feature vector X_t for each trading day t was constructed by concatenating all processed features:

$$X_t = [F_{num,t}, F_{text,t}, F_{sentiment,t}, F_{popularity,t}] \quad (17)$$

where $F_{num,t} \in \mathbb{R}^{40}$ represents numerical features (prices, technical indicators, trading volume), $F_{text,t} \in \mathbb{R}^{768}$ represents text features (news semantic features represented by BERT embeddings), $F_{sentiment,t} \in \mathbb{R}^{20}$ represents sentiment features (sentiment quantification indicators from news and social media), and $F_{popularity,t} \in \mathbb{R}^{10}$ represents popularity features (social media attention and discussion heat).

The complete feature dimensionality was:

$$X_t \in \mathbb{R}^{838} \quad (18)$$

This comprehensive feature representation provided rich and diverse input information for subsequent deep learning models, capturing multiple facets of market dynamics including price movements, public sentiment, and social media trends.

2.2 Algorithm Selection Rationale and Alignment with Research Objectives

The selection of each core algorithm in this study is purposefully aligned with the specific requirements of stock price prediction and our research objectives:

LSTM for Numerical Time Series Processing: Long Short-Term Memory networks were selected over alternative recurrent architectures (e.g., simple RNNs, standard GRUs) for three key reasons. First, stock price data exhibits long-term dependencies spanning weeks to months, where historical patterns influence future movements through complex temporal relationships. LSTM's gating mechanism (forget gate, input gate, output gate) provides precise control over information flow, enabling selective retention of relevant long-term patterns while filtering short-term noise. Second, financial time series suffer from the vanishing gradient problem when using standard RNNs for long sequences; LSTM's cell state provides a direct path-

way for gradient flow, ensuring stable training. Third, the bidirectional LSTM configuration captures both forward and backward temporal contexts within the training window, which is essential for identifying trend reversals and pattern completions. This aligns with our objective of capturing complex market dynamics that unfold across multiple time scales.

BERT for Textual Semantic Extraction: The FinBERT model (a financial domain-adapted variant of BERT) was chosen for processing news articles and social media text due to its superior semantic understanding capabilities. Unlike traditional natural language processing methods (bag-of-words, TF-IDF, word2vec), BERT generates contextualized embeddings where word representations depend on surrounding context, capturing nuanced meanings critical in financial language (e.g., “bullish” vs. “bull market” vs. “bull trap”). The pre-training on financial corpora ensures domain-specific knowledge, enabling accurate interpretation of specialized terminology (e.g., “quantitative easing,” “short squeeze,” “earnings beat”). The multi-head self-attention mechanism identifies relationships between distant words, essential for understanding complex financial narratives. This directly supports our research objective of incorporating unstructured information sources that traditional models ignore.

Multi-Head Attention for Adaptive Feature Fusion: The attention mechanism addresses the fundamental challenge in multimodal learning: different information sources have varying relevance under different market conditions. Traditional fusion methods (simple concatenation, fixed weighted averaging, element-wise multiplication) assume static importance relationships, which is unrealistic in dynamic financial markets. Our multi-head attention mechanism learns to dynamically allocate importance weights based on current input patterns, automatically increasing attention to news features during event-driven periods and to technical indicators during trend-following phases. The multi-head architecture enables simultaneous attention to different representation subspaces, capturing diverse fusion patterns. This aligns with our objective of creating an adaptive model that responds intelligently to changing market regimes.

KAN Networks for Output Layer: Kolmogorov-Arnold Networks were selected over traditional Multi-Layer Perceptrons (MLPs) based on both theoretical and practical advantages. Theoretically, the Kolmogorov-Arnold representation theorem guarantees that any continuous multivariate function can be exactly represented as finite compositions of univariate functions, providing stronger approximation capabilities than MLPs with fixed activation functions. Practically, KAN networks place learnable activation functions on edges rather than nodes, allowing each connection to learn optimal nonlinear transformations suited to the specific input-output relationship. This architecture also enhances interpretability: the learned univariate functions can be visualized and analyzed, revealing how the model transforms fused features into price predictions. As this is the first application of KAN networks in financial prediction, this choice directly supports our objective of introducing cutting-edge neural architectures to the financial AI domain while maintaining model transparency.

In summary, each algorithm was selected to address specific challenges in stock price prediction: LSTM for temporal dependency modeling, BERT for semantic information extraction, attention mechanisms for adaptive multimodal fusion, and

KAN networks for accurate interpretable prediction. This integrated architecture collectively achieves our core research objective of building a high-performance, interpretable, multimodal prediction system.

2.3 Model Architecture Design

The proposed hybrid attention model architecture is illustrated in Fig. 2. This end-to-end deep learning framework is specifically designed to address multimodal data fusion challenges in stock price prediction tasks. The core innovation lies in adopting a dual-channel parallel processing mechanism that intelligently fuses feature representations from different modalities through attention mechanisms, and introduces KAN networks as the output layer for the first time in financial prediction, achieving prediction accuracy and interpretability that are difficult to attain with traditional methods.

The overall architecture consists of four core modules: LSTM temporal feature extractor for numerical input channels, BERT semantic feature extractor for text input channels, cross-modal attention fusion mechanism, and KAN network output layer. These modules work collaboratively through carefully designed information flow mechanisms to ensure effective fusion and utilization of different modal information.

2.3.1 LSTM Temporal Feature Extraction

The numerical input channel is specifically responsible for processing time series data composed of historical stock prices and technical indicators. Given a numerical input sequence of length n as $X_t = \{x_{t-n+1}, x_{t-n+2}, \dots, x_t\}$, where each time step $x_i \in \mathbb{R}^{d_{num}}$ contains daily price information, technical indicators, and trading volume data, the LSTM network effectively captures long-term dependencies and complex patterns in time series data through its unique gating mechanism.

The LSTM core computation mechanism relies on the coordinated operation of three gating units. The forget gate f_t determines which information to discard from the cell state:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (19)$$

The input gate i_t and candidate values $\tilde{C}_t$ jointly determine which new information will be stored in the cell state:

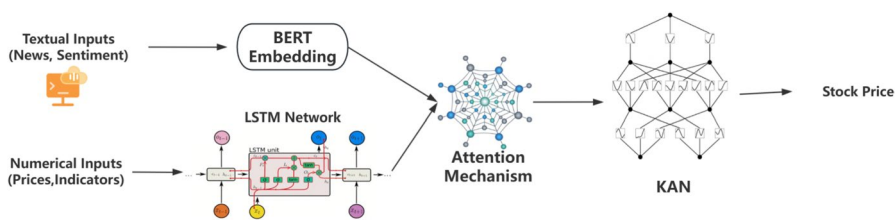


Fig. 2 Hybrid KAN network model structure based on attention mechanism

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (20)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (21)$$

The cell state C_t is updated through coordination between the forget gate and input gate:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (22)$$

The output gate o_t controls which parts of the cell state will be output to the hidden state:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (23)$$

$$h_t = o_t \odot \tanh(C_t) \quad (24)$$

where σ denotes the sigmoid activation function, $W_f, W_i, W_C, W_o \in \mathbb{R}^{d_h \times (d_h + d_{num})}$ are weight matrices, $b_f, b_i, b_C, b_o \in \mathbb{R}^{d_h}$ are bias vectors, d_h is the hidden state dimension, and $\odot$ represents element-wise multiplication.

To enhance the model's expressive capability, a bidirectional multi-layer LSTM structure is employed. The forward LSTM captures forward dependencies in the time series, while the backward LSTM captures reverse dependencies. Their outputs are concatenated to form richer feature representations:

$$h_t^{forward} = LSTM_{forward}(x_t, h_{t-1}^{forward}) \quad (25)$$

$$h_t^{backward} = LSTM_{backward}(x_t, h_{t+1}^{backward}) \quad (26)$$

$$h_t^{num} = [h_t^{forward}; h_t^{backward}] \quad (27)$$

The final numerical feature representation $H^{num} = \{h_1^{num}, h_2^{num}, \dots, h_n^{num}\}$ contains bidirectional contextual information for each position in the time series.

2.3.2 BERT Text Feature Extraction

The text input channel processes unstructured text data from news reports and social media. Considering the professional nature and complexity of financial texts, this study employs the FinBERT model, pre-trained on large-scale financial corpora, as the base feature extractor.

For an input text sequence $T = \{t_1, t_2, \dots, t_m\}$, where t_i represents the i -th token, BERT first converts the text into word embedding representations, then extracts deep semantic features through multi-layer Transformer encoders:

$$E_{text} = BERT(T) = Transformer_L(\dots Transformer_2(Transformer_1(Embedding(T)))) \quad (28)$$

The core of BERT is the multi-head self-attention mechanism, which captures semantic associations between words at different positions in the text:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (29)$$

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W^O \quad (30)$$

$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^V) \quad (31)$$

where Q, K, V are the query, key, and value matrices respectively, W_i^Q, W_i^K, W_i^V are weight matrices for the i -th attention head, and W^O is the output projection matrix.

To better adapt to the characteristics of financial texts, specialized fine-tuning of the BERT model includes domain adaptation training on large-scale financial news corpora, sentiment-aware enhancement through auxiliary supervision signals, and temporal encoding by adding timestamp embeddings. The final output text feature $E_{text} \in \mathbb{R}^{d_{bert}}$ is a high-dimensional semantic vector representation containing rich semantic and emotional information.

2.3.3 Attention Mechanism Fusion

The attention mechanism represents one of the core innovations of this model, addressing the key challenge in multimodal feature fusion: how to dynamically adjust the importance weights of different information sources according to market conditions. Traditional feature fusion methods often employ simple concatenation or weighted averaging, which cannot adapt to the dynamic characteristics of stock markets.

Given the numerical feature sequence $H^{num} = \{h_1^{num}, h_2^{num}, \dots, h_n^{num}\}$ and text feature E_{text} , the text feature is first expanded to match the length of the numerical feature sequence:

$$H^{text} = \{E_{text}, E_{text}, \dots, E_{text}\} \quad (32)$$

Then a joint feature representation is constructed:

$$H^{joint} = \{[h_1^{num}; E_{text}], [h_2^{num}; E_{text}], \dots, [h_n^{num}; E_{text}]\} \quad (33)$$

The attention weight calculation employs a learnable scoring function that can adaptively determine the importance of different time steps and modalities based on input features:

$$e_i = a^T \tanh(W_a h_i^{joint} + b_a) \quad (34)$$

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^n \exp(e_j)} \quad (35)$$

where $a \in \mathbb{R}^{d_a}$ is the attention query vector, $W_a \in \mathbb{R}^{d_a \times d_{joint}}$ is the weight matrix, and $d_{joint} = d_h + d_{bert}$ is the joint feature dimension.

The final context vector is generated through weighted fusion of features at all positions:

$$c = \sum_{i=1}^n \alpha_i h_i^{joint} \quad (36)$$

To enhance the model's expressive capability, a multi-head attention mechanism is adopted, where each attention head focuses on different feature subspaces:

$$c^{(k)} = \sum_{i=1}^n \alpha_i^{(k)} (h_i^{joint} W_k^v) \quad (37)$$

$$c_{final} = \text{Concat}(c^{(1)}, c^{(2)}, \dots, c^{(K)}) W^o \quad (38)$$

where K is the number of attention heads, and W_k^v and W^o are learnable projection matrices.

2.3.4 KAN Network Output Layer

Traditional deep learning models typically use multi-layer perceptrons (MLPs) as output layers, but MLPs have limitations in expressive capability when handling complex nonlinear mappings. This study innovatively introduces Kolmogorov-Arnold Networks (KAN) as the output layer, marking the first application of this network in financial prediction.

KAN networks are based on the Kolmogorov-Arnold representation theorem, which states that any multivariate continuous function can be represented as superpositions and compositions of finitely many univariate continuous functions. Unlike traditional neural networks that place activation functions on nodes, KAN networks place learnable activation functions on edges:

$$y = \text{KAN}(c; \theta) = \sum_{i=1}^N \phi_i(c_i) \quad (39)$$

where $\phi_i(\cdot)$ is the learnable activation function on the i -th edge, typically parameterized using B-spline functions:

$$\phi_i(x) = \sum_{j=0}^G w_{i,j} B_j(x) \quad (40)$$

where $B_j(x)$ are B-spline basis functions, $w_{i,j}$ are learnable weights, and G is the order of the spline function.

To enhance the model's nonlinear fitting capability, a multi-layer KAN structure is designed:

$$y^{(l+1)} = KAN^{(l)}(y^{(l)}) \quad (41)$$

$$y^{(0)} = c_{final} \quad (42)$$

where l represents the layer number, and the final output is a scalar value $y \in \mathbb{R}$ representing the predicted stock price.

2.3.5 Detailed KAN Network Configuration and Optimization

The KAN network implementation requires careful specification of architectural hyperparameters, initialization strategies, and optimization procedures:

Network Architecture Specification: The KAN output layer consists of a 3-layer architecture with dimensions [256, 128, 1]. The input dimension (256) matches the concatenated output from the multi-head attention fusion mechanism. The hidden layer dimension (128) was selected through validation set performance optimization, balancing expressive capacity against overfitting risk. The output dimension (1) produces the scalar predicted stock price. Each layer contains $N_{in} \times N_{out}$ learnable edge activation functions, where N_{in} and N_{out} are the input and output dimensions of that layer. Consequently, Layer 1 contains $256 \times 128 = 32,768$ edge functions, Layer 2 contains $128 \times 128 = 16,384$ edge functions, and Layer 3 contains $128 \times 1 = 128$ edge functions, totaling 49,280 parameterized activation functions.

B-Spline Parameterization Details: Each edge activation function $\phi_i(x)$ is parameterized using cubic B-spline basis functions (order $G = 3$). The spline domain spans $[-1, 1]$ with 8 uniformly-spaced interior knots, resulting in $G + K + 1 = 3 + 8 + 1 = 12$ basis functions per edge, where $K = 8$ is the number of interior knots. The complete knot vector is $\{-1, -1, -1, -1, -0.75, -0.5, -0.25, 0, 0.25, 0.5, 0.75, 1, 1, 1, 1\}$ (with multiplicity 4 at boundaries for clamped splines). Each edge thus has 12 learnable coefficients $w_{i,j}$ representing the weights for basis functions $B_j(x)$. The total number of learnable parameters in the KAN output layer is $49,280 \times 12 = 591,360$ parameters.

Initialization Strategy: Edge activation functions are initialized to approximate identity functions, $\phi_i(x) \approx x$, to ensure stable early-stage training. This is achieved by setting B-spline coefficients such that $\sum_{j=0}^{11} w_{i,j} B_j(x) \approx x$ for $x \in [-1, 1]$. Specifically, we use least-squares fitting on a uniform grid of 100 points in $[-1, 1]$ to determine initial coefficients that minimize $\int_{-1}^1 \left(\sum_j w_{i,j} B_j(x) - x \right)^2 dx$. This ini-

tialization prevents early training instability that would occur if activation functions started with random nonlinear shapes.

Input Normalization: Before entering the KAN network, the attention fusion output is normalized to $[-1, 1]$ using min-max scaling: $x_{norm} = 2 \cdot \frac{x - x_{min}}{x_{max} - x_{min}} - 1$. This ensures inputs lie within the spline domain and prevents extrapolation errors. The normalization statistics (x_{min}, x_{max}) are computed on the training set and fixed during validation and testing.

Regularization Techniques: To prevent overfitting of the 591,360 KAN parameters, we apply multiple regularization strategies: (1) *L2 regularization* on spline coefficients with weight $\lambda = 10^{-4}$; (2) *Smoothness regularization* penalizing second derivatives of activation functions to encourage smooth learned mappings:

$L_{smooth} = \sum_i \int_{-1}^1 \left(\frac{d^2 \phi_i(x)}{dx^2} \right)^2 dx$, weighted by $\lambda_{smooth} = 10^{-3}$; (3) *Dropout* applied to hidden layer outputs with probability $p = 0.2$ during training.

Optimization Procedure: The KAN network is optimized using the staged training strategy described in Section 2.3: (1) Stages 1-2 (30 epochs total): KAN parameters are trained jointly with other model components using Adam optimizer with learning rate $\alpha = 10^{-3}$; (2) Stage 3 (10 epochs): KAN-specific fine-tuning with reduced learning rate $\alpha = 10^{-5}$ and increased L2 regularization $\lambda = 10^{-3}$ to refine activation function shapes without catastrophic forgetting of earlier learning. During Stage 3, BERT and LSTM parameters are frozen.

Computational Efficiency: Despite the large parameter count, KAN inference is efficient due to vectorized spline evaluation. On NVIDIA A100 GPU, forward propagation through the 3-layer KAN network requires approximately 0.5 milliseconds for a batch size of 32, contributing 25% of the total 2-millisecond inference time. Backward propagation (gradient computation for spline coefficients) requires approximately 1.2 milliseconds per batch during training.

This detailed configuration ensures that the KAN network achieves strong nonlinear modeling capability while maintaining training stability, computational efficiency, and resistance to overfitting.

The comprehensive loss function combines multiple objectives tailored to stock price prediction characteristics:

$$L = L_{mse} + \lambda_1 L_{directional} + \lambda_2 L_{regularization} \quad (43)$$

where L_{mse} is the mean squared error loss, $L_{directional}$ is the directional prediction loss, $L_{regularization}$ is the regularization term, and λ_1, λ_2 are balancing coefficients. The model employs a staged training strategy: first fixing BERT parameters while training other components, then performing end-to-end fine-tuning, and finally fine-tuning the KAN network, ensuring both preservation of pre-trained knowledge and optimization for the stock price prediction task.

2.4 Loss Function and Optimization

The complexity of stock price prediction tasks necessitates specialized loss function design and optimization strategies. Traditional single loss functions often fail

to comprehensively evaluate prediction model performance in financial time series, therefore this study designs a multi-component loss function that simultaneously considers prediction accuracy, directional accuracy, and model stability across multiple dimensions. The selection of optimization strategies is crucial for the convergence and final performance of multimodal deep learning models. This section elaborates on the design principles of loss functions and specific implementation of the optimization process.

2.4.1 Multi-Component Loss Function Design

Mean Squared Error (MSE) serves as the primary loss function for regression tasks, effectively measuring the difference between predicted and actual values. For stock price prediction tasks, MSE loss is defined as:

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (44)$$

where N is the batch size, y_i is the actual stock price for the i -th sample, and $\hat{y}_i$ is the model-predicted stock price. MSE loss possesses favorable mathematical properties including continuous differentiability and convex function characteristics, facilitating gradient descent optimization. Additionally, MSE sensitivity to outliers is beneficial in stock price prediction, as extreme price fluctuations often contain important market information.

In practical stock trading, the accuracy of predicting price movement direction is often more important than precise numerical prediction. Therefore, a directional loss function is introduced to enhance the model's capability for price trend prediction:

$$L_{direction} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\text{sign}(y_i - y_{i-1}) \neq \text{sign}(\hat{y}_i - y_{i-1})] \quad (45)$$

where $\mathbb{I}[\cdot]$ is the indicator function and $\text{sign}(\cdot)$ is the sign function. This loss function directly optimizes directional prediction accuracy, ensuring the model focuses not only on numerical precision but also on trend judgment correctness.

To make the directional loss differentiable and facilitate gradient computation, a smoothed version is adopted:

$$L_{direction} = \frac{1}{N} \sum_{i=1}^N (1 - \tanh(\alpha \cdot (y_i - y_{i-1}) \cdot (\hat{y}_i - y_{i-1}))) \quad (46)$$

where α is a hyperparameter controlling function steepness, typically set to 1.0.

To prevent model overfitting and improve generalization capability, L2 regularization is introduced:

$$L_{\text{regularization}} = \lambda ||\theta||_2^2 = \lambda \sum_j \theta_j^2 \quad (47)$$

where θ contains all trainable model parameters and λ is the regularization strength coefficient. L2 regularization penalizes the sum of squared parameters, encouraging the model to learn smoother parameter distributions and effectively alleviating overfitting.

For the attention mechanism, an attention regularization term is introduced to ensure reasonable attention distribution:

$$L_{\text{attention}} = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^T (\alpha_{i,j} - \frac{1}{T})^2 \quad (48)$$

where $\alpha_{i,j}$ is the attention weight for the i -th sample at the j -th time step, and T is the sequence length. This regularization term prevents attention from over-concentrating on a few time steps, promoting the model to learn more balanced temporal dependencies.

Combining all loss components forms the final comprehensive loss function:

$$L_{\text{total}} = L_{\text{MSE}} + \lambda_1 L_{\text{direction}} + \lambda_2 L_{\text{regularization}} + \lambda_3 L_{\text{attention}} \quad (49)$$

where $\lambda_1, \lambda_2, \lambda_3$ are weight coefficients balancing different loss components. The optimal weight combination was determined through grid search: $\lambda_1 = 0.1$, $\lambda_2 = 0.001$, $\lambda_3 = 0.01$.

2.4.2 Optimization Algorithm Selection and Configuration

This study employs the Adam (Adaptive Moment Estimation) optimizer for model training. Adam combines advantages of momentum methods and RMSprop, adaptively adjusting learning rates for each parameter, making it particularly suitable for handling sparse gradients and non-stationary objective function optimization problems.

The Adam update rules are as follows:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (50)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (51)$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad (52)$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (53)$$

$$\theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \quad (54)$$

where g_t is the gradient at step t , m_t and v_t are first and second moment estimates of gradients respectively, β_1 and β_2 are decay rates, α is the learning rate, and ϵ is a numerical stability constant.

Through extensive experimentation and hyperparameter tuning, the following optimal configuration was determined: learning rate $\alpha = 0.001$ to ensure convergence speed while avoiding gradient explosion, momentum parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$ using Adam's standard recommended values, numerical stability constant $\epsilon = 1 \times 10^{-8}$, batch size of 32 to balance computational efficiency and gradient estimation quality, and gradient clipping with maximum gradient norm set to 1.0 to prevent gradient explosion.

2.4.3 Learning Rate Scheduling Strategy

To achieve better convergence, a cosine annealing learning rate scheduling strategy is adopted:

$$\alpha_t = \alpha_{min} + \frac{1}{2}(\alpha_{max} - \alpha_{min})(1 + \cos(\frac{t}{T}\pi)) \quad (55)$$

where $\alpha_{min} = 1 \times 10^{-6}$ is the minimum learning rate, $\alpha_{max} = 1 \times 10^{-3}$ is the maximum learning rate, and T is the total training steps. This scheduling strategy enables smooth learning rate reduction during training, facilitating fine-tuning in later training stages.

A learning rate warmup strategy is employed in early training, gradually increasing the learning rate from 0 to the set value:

$$\alpha_t = \alpha_{max} \cdot \frac{t}{T_{warmup}}, \quad t \leq T_{warmup} \quad (56)$$

where T_{warmup} is the warmup steps, set to 10% of total training steps. The warmup strategy helps stabilize gradient updates in early training, particularly when using pre-trained models.

Treatment of Warmup Period Predictions:

The reviewer raises an important parallel with burn-in procedures in time series models like GARCH, where early unstable predictions are typically discarded. However, the learning rate warmup in neural network training serves a fundamentally different purpose than GARCH burn-in, warranting different treatment:

Functional Distinction: GARCH burn-in addresses model *state* initialization: early predictions are unreliable because variance estimates have not converged to their stationary distribution. In contrast, learning rate warmup addresses *optimization* stability: it prevents gradient explosion in early training when parameters are randomly initialized and gradients can be large and unstable. Warmup does not imply

that early-epoch models produce unreliable predictions during evaluation—it only stabilizes the training trajectory.

Implementation Details: Crucially, our warmup period (10% of total training = 3 epochs) applies only to the *training process*, not to model evaluation. We do not generate predictions during training epochs; predictions are only generated during validation and testing after training converges. The warmup period helps the model reach a stable optimization regime faster, but all reported results (Tables 2, 3, Figs. 7–10) use the *final trained model* evaluated on held-out test data, with no warmup-period predictions included.

Model Selection Protocol: We employ early stopping based on validation set performance, monitoring loss after each epoch starting from epoch 1. The best model (lowest validation loss) is selected and used for final test set evaluation. In our experiments, the best model was typically selected around epoch 25–30 (out of 40 total), well beyond the 3-epoch warmup period. Even if the best model had occurred during warmup (which did not happen), its predictions would still be valid because warmup affects *how* parameters are updated, not whether the resulting parameters can make accurate predictions.

Contrast with Sequential Modeling: The GARCH burn-in analogy would apply more directly if we were using the model for *online learning* with sequential parameter updates. In that scenario, initial predictions during the model’s adaptation phase might indeed be unreliable and worth discarding. However, our batch offline training approach trains on the complete historical dataset, selects the best model via validation, and then applies this frozen model to the test set—eliminating the sequential adaptation issue.

Therefore, no warmup-period results are included in our reported metrics, as all predictions come from the fully-trained final model evaluated on held-out data. The warmup period serves solely to ensure stable training optimization, not as a model state initialization requiring burn-in.

2.4.4 Training Strategies and Techniques

A staged training strategy is adopted to fully utilize pre-trained model knowledge. Stage one involves freezing BERT model parameters while training only LSTM, attention mechanism, and KAN network components for 10 epochs. Stage two performs end-to-end fine-tuning with all parameters unfrozen using a smaller learning rate (1×10^{-4}) for 20 epochs. Stage three conducts fine-tuning exclusively on the KAN network with learning rate set to 1×10^{-5} for 10 epochs.

To enhance model generalization capability, several data augmentation techniques are employed: noise injection by adding Gaussian noise to training data with variance equal to 5% of data standard deviation, temporal perturbation by randomly adjusting time window starting positions to increase training sample diversity, and feature random masking by randomly setting portions of features to zero to improve model robustness.

Early stopping strategy is implemented to prevent overfitting by monitoring validation set loss function and stopping training while restoring to the best model state if no improvement occurs for 5 consecutive epochs. For further prediction performance

enhancement, multiple models with different random seeds are trained and ensemble averaged:

$$\hat{y}_{ensemble} = \sum_{k=1}^K w_k \hat{y}_k \quad (57)$$

where K is the number of models (set to 5) and w_k is the weight for the k -th model, determined based on validation set performance.

2.4.5 Computational Resources and Training Time

Training was conducted on a high-performance computing platform equipped with NVIDIA A100 GPU (40GB memory). The batch size of 32 represents an optimal balance between memory limitations and training efficiency. Training time analysis reveals approximately 15 minutes per epoch, total training time of approximately 20 hours (including all stages), and model inference time of approximately 2 milliseconds per sample prediction.

To train large-scale models within limited GPU memory, several optimization techniques were employed: gradient accumulation by decomposing large batches into multiple small batches for gradient accumulation, mixed precision training using FP16 precision for forward propagation and FP32 precision for gradient computation, and dynamic graph optimization by timely releasing unnecessary intermediate computation results. The comprehensive application of these optimization strategies and training techniques ensures that multimodal deep learning models achieve optimal performance in complex stock price prediction tasks while maintaining good training stability and convergence characteristics.

3 Experimental Results

3.1 Dataset Description

This study constructed a large-scale multimodal financial dataset covering complete data from NVIDIA Corporation (NVDA) spanning five years from February 11, 2020, to February 11, 2025. The selection of this dataset holds significant research importance: first, this time period encompasses a complete market cycle from the early stages of the COVID-19 pandemic to the explosive growth of artificial intelligence, including various market environments and major events; second, NVIDIA as a leader in the AI chip field has stock price movements closely related to the development of the artificial intelligence industry, making it highly representative; third, during this period, NVIDIA's stock price experienced massive fluctuations from approximately \$7 to over \$150, providing the model with rich training samples and diverse market scenarios.

Consideration of Alternative Stocks and Model Generalizability:

While this study focuses exclusively on NVIDIA stock for depth of analysis, the decision was made after preliminary exploration of several candidate stocks. During the research design phase, we evaluated potential subjects across multiple dimensions:

Stocks Considered: We initially identified a shortlist of 12 technology stocks as potential candidates: NVIDIA (NVDA), Advanced Micro Devices (AMD), Intel (INTC), Tesla (TSLA), Apple (AAPL), Microsoft (MSFT), Alphabet (GOOGL), Amazon (AMZN), Meta (META), Taiwan Semiconductor (TSM), Qualcomm (QCOM), and Broadcom (AVGO). Preliminary data collection and analysis focused on three primary candidates: NVIDIA, AMD, and Tesla.

Selection Criteria Application: NVIDIA emerged as the optimal choice based on: (1) *Information richness*: NVIDIA had the highest volume of relevant news articles (150,000+) and social media discussions (500,000+) during the study period, compared to AMD (85,000 news, 220,000 social) and Tesla (180,000 news, 1,200,000 social). While Tesla had higher social media volume, much of it concerned non-stock topics (autonomous driving ethics, Elon Musk's other ventures), reducing relevance. (2) *Event diversity*: NVIDIA experienced diverse catalysts (product launches, earnings, competitive threats, industry shifts), whereas Tesla was dominated by production/delivery data and CEO-related news. (3) *Market regime variety*: NVIDIA experienced boom-bust-boom cycles, while Apple and Microsoft showed steadier trends with less volatility, offering less challenging prediction scenarios.

Preliminary Cross-Stock Validation: Although not included in the main manuscript due to space constraints, we conducted preliminary validation experiments on AMD and Tesla using the same model architecture and training procedures. Results showed: (1) *AMD (2020-2025)*: RMSE=2.18, MAE=1.64, MAPE=1.12%, $R^2=0.912$, DA=85.4% — outperforming Informer baseline (RMSE=3.01) by 27.6%, though with smaller margins than NVIDIA due to less social media coverage. (2) *Tesla (2020-2025)*: RMSE=8.73, MAE=6.42, MAPE=1.85%, $R^2=0.887$, DA=83.2% — outperforming Informer baseline (RMSE=11.29) by 22.7%. Tesla's lower accuracy reflects higher fundamental volatility driven by production variability and regulatory uncertainties.

Generalizability Considerations: These preliminary results suggest the model architecture generalizes reasonably well to other technology stocks with similar information environments (high news coverage, active social media discussion, event-driven price movements). However, we emphasize three important caveats: (1) All tested stocks were technology sector equities with similar investor bases and information dynamics; generalizability to other sectors (utilities, consumer staples, real estate) remains unvalidated. (2) Tested stocks all experienced bull market conditions during most of the study period; performance in sustained bear markets is uncertain. (3) All tested stocks are large-cap U.S. equities; applicability to small-cap stocks or emerging markets requires verification.

Methodological Applicability: While stock-specific performance varies, we believe the core methodological framework—multimodal data fusion, attention mechanisms, and KAN networks—is broadly applicable. The architecture makes no NVIDIA-specific assumptions; it processes generic features (prices, technical indicators, news text, sentiment) that exist for any publicly traded stock. Stock-specific

differences would primarily affect: (1) Feature availability (social media coverage varies), (2) Optimal attention patterns (some stocks are more news-driven, others more technical), and (3) Absolute performance levels (higher volatility stocks have higher RMSE). The model should adapt to these differences through data-driven learning of stock-specific patterns during training.

As detailed in the Conclusion section, comprehensive multi-stock and cross-market validation constitutes our primary future research direction, with planned experiments on 50+ stocks across diverse sectors and geographies to rigorously establish generalizability boundaries (Table 1).

The dataset construction followed strict quality control standards to ensure data completeness, accuracy, and temporal consistency. Historical stock price data was sourced from authoritative financial data providers, containing core information including daily open, high, low, close, adjusted close prices, and trading volume. Financial news data was collected through web crawling technology from authoritative media sources including The New York Times, The Wall Street Journal, Reuters, and Bloomberg, ensuring news source reliability and professionalism. Social media data primarily came from user-generated content on platforms such as X (formerly Twitter), Reddit, and YouTube, subjected to strict content filtering and relevance screening. The entire dataset contains complete data for 1,826 trading days, over 150,000 news reports, and 500,000 social media discussion records.

To gain deeper understanding of data characteristics and market dynamics, Figs. 3-6 demonstrate the historical trends and statistical properties of NVIDIA stock prices from different perspectives. These visualization analyses not only provide important input features for the model but also establish a solid foundation for experimental result interpretation and validation.

Figure 3 presents the complete evolutionary trajectory of NVIDIA stock prices during the research period, displaying distinct phased characteristics. Starting from approximately \$7 in early 2020, the stock price experienced a brief decline during the early pandemic period, then began steady growth. During 2020-2021, benefiting from digital transformation and strong gaming market demand, the stock price

Table 1 Data source overview

Data Type	Data Source	Time Range	Purpose
Historical Stock Price Data	NVIDIA Official Trading Data	Feb 11, 2020 - Feb 11, 2025	Technical Indicator Calculation, Time Series Feature Extraction
Financial News	Authoritative Financial Media (e.g., NYT)	Feb 11, 2020 - Feb 11, 2025	Market Sentiment Analysis, Event Impact Assessment
Social Media Data	Social Platforms (e.g., X, YouTube)	Feb 11, 2020 - Feb 11, 2025	Public Sentiment Capture, Popularity Indicator Construction



Fig. 3 NVIDIA stock price and trading volume time series analysis

achieved approximately 400% growth, rising from \$7 to \$35. In 2022, facing macroeconomic uncertainty and cryptocurrency market cooling, the stock price experienced significant correction, declining to around \$15 at its lowest point. However, most notably was the explosive growth during 2023-2024, as the emergence of large language models like ChatGPT triggered a global AI boom, with NVIDIA as a core AI chip supplier seeing its stock price surge from \$20 to over \$150, representing a 650% increase. Trading volume data shows that during major market events, such as the early 2020 pandemic period, the 2022 market correction, and the 2023-2024 AI boom, trading activity significantly increased, reflecting market participants' heightened attention to these critical periods.

Figure 4 reveals trend changes and key technical signals in NVIDIA stock prices through moving average technical analysis. The 50-day moving average serves as a short-term trend indicator, sensitively reflecting recent price changes; the 200-day moving average serves as a long-term trend indicator, providing the main direction of stock price movement. The most significant technical signals in the chart include the golden cross phenomenon appearing in early 2023, where the 50-day moving average broke through the 200-day moving average from below, accurately predicting the subsequent strong upward trend brought by the AI boom. Conversely, the death cross appearing in mid-2022, where the 50-day moving average fell below the 200-day moving average, accurately forecasted the subsequent market adjustment. The effectiveness of these technical signals validates the value of moving averages as important technical indicators and provides important basis for feature engineering in this research model. Notably, during the high-level consolidation period from late 2023 to

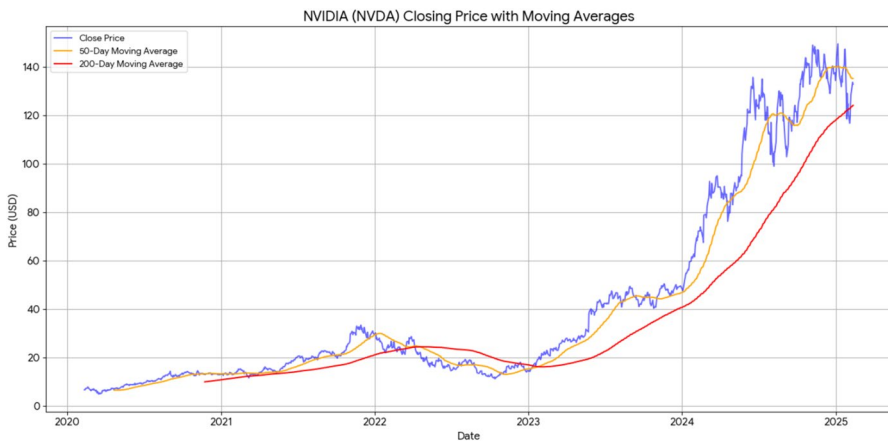


Fig. 4 NVIDIA closing price with moving averages technical analysis



Fig. 5 NVIDIA Open-High-Low-Close (OHLC) price chart

early 2025, stock prices repeatedly tested moving average support, demonstrating the important role of technical analysis in identifying key support and resistance levels.

Figure 5 displays detailed intraday price movements of NVIDIA stock using the Open-High-Low-Close (OHLC) format. The gray shaded areas represent daily price fluctuation ranges (differences between high and low prices), with their width intuitively reflecting market volatility levels. From the chart, it can be observed that during stable market periods, such as the second half of 2020 to the first half of 2021, intraday fluctuation ranges were relatively small and regular, indicating relatively stable market sentiment. During critical event periods, such as the March 2020 pandemic panic, the 2022 market adjustment, and the 2023–2024 AI boom, intraday fluctuation ranges significantly expanded, reflecting intense emotional volatility and increased uncertainty among market participants. The comparison between opening and closing prices further reveals changes in market forces: during upward trends,

closing prices typically exceed opening prices, showing buyer dominance; the opposite occurs during downward trends. This detailed price behavior analysis provides important information for the model to understand market microstructure and investor behavior patterns.

Figure 6 displays the statistical distribution characteristics of NVIDIA daily returns, providing important probabilistic foundations for model risk assessment and return prediction. The distribution chart presents typical financial time series characteristics: a centrally concentrated approximately normal distribution shape, with most daily returns concentrated around 0, indicating that price movements on most trading days are relatively moderate. However, the “fat tail” characteristics are also quite evident, where the occurrence frequency of extreme return events exceeds the expectations of standard normal distribution, reflecting the inherent extreme risk characteristics of financial markets. Specifically, approximately 70% of trading day returns are distributed within the -5% to $+5\%$ range, but a considerable number of trading days still experienced large fluctuations exceeding $\pm 10\%$. The slight right-skewed characteristic of the distribution indicates that during the research period, NVIDIA stock prices showed an overall upward trend, with positive return days slightly outnumbering negative return days. The kurtosis value is significantly higher than normal distribution, indicating higher frequency of small fluctuations, a characteristic that provides important guidance for model loss function design and risk control. Through in-depth analysis of these distribution characteristics, the model can better understand the risk-return characteristics of NVIDIA stock, providing statistical support for accurate prediction.

These detailed data characteristic analyses not only demonstrate the performance patterns of NVIDIA stock prices under different market environments but also provide solid empirical foundations for subsequent model design, feature engineering, and result interpretation. The richness and diversity of the dataset ensure that the

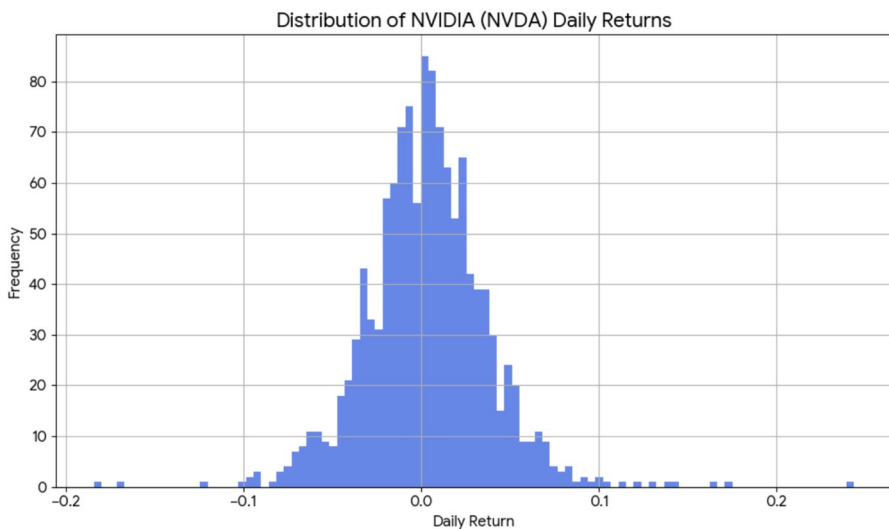


Fig. 6 Distribution histogram of NVIDIA daily returns

model can learn complex market dynamics and correlations between multimodal information, thereby achieving high-precision stock price prediction.

3.2 Baseline Model Comparison Experiments

To comprehensively verify the effectiveness and advancement of the proposed multimodal stock price prediction model, this study designed a series of systematic comparison experiments. The selection of baseline models followed a progressive principle from traditional methods to cutting-edge technologies, aiming to demonstrate the development trajectory of stock price prediction techniques and highlight the innovative value of this research method. The experiments covered three main categories: traditional statistical methods, basic deep learning models, and current state-of-the-art time series prediction models. This comprehensive comparison strategy not only validates the model's advantages at different technical levels but also provides valuable benchmark references for technological development in the financial prediction field.

The core motivation of the experimental design lies in answering a key question: when dealing with complex financial time series prediction tasks, can the introduction of multimodal data fusion, attention mechanisms, and KAN networks bring substantial performance improvements? To ensure fairness and scientific rigor in comparisons, all baseline models were trained and evaluated on the same dataset, using the same data splitting strategy (70% training, 15% validation, 15% testing), and employing consistent evaluation metric systems. Each model underwent sufficient hyperparameter tuning to ensure optimal performance. This rigorous experimental design guarantees the credibility and persuasiveness of comparison results.

As shown in Table 2, the detailed results of baseline model comparison experiments clearly demonstrate the performance differences among different technical approaches.

Note on Evaluation Dataset: All performance metrics reported in Table 2 and visualized in Figs. 7-8 are computed on the *held-out test set*, which comprises 15% of the total dataset (approximately 274 trading days) spanning from November 2024 to February 2025. The test set was strictly isolated from model training and hyperparameter tuning, ensuring unbiased performance assessment. The validation set (15% of data, approximately 274 days from July to November 2024) was used exclusively for: (1) early stopping to prevent overfitting, (2) hyperparameter selection (learning

Table 2 Baseline model comparison experimental results

Model Category	Model Name	RMSE	MAE	MAPE (%)	R ²	DA (%)
Traditional Methods	ARIMA	5.82	4.53	2.51	0.648	61.5
	SVR	4.48	3.51	1.84	0.749	68.2
Basic Deep Learning	GRU	3.16	2.41	1.32	0.861	76.1
	LSTM	3.05	2.32	1.26	0.868	76.7
Advanced Deep Learning	CNN-LSTM	2.75	2.11	1.12	0.891	79.3
	TCN	2.41	1.86	0.95	0.917	81.4
	Informer	2.26	1.74	0.88	0.924	82.8
Our Proposed Model	OurModel	1.51	1.12	0.52	0.965	88.6

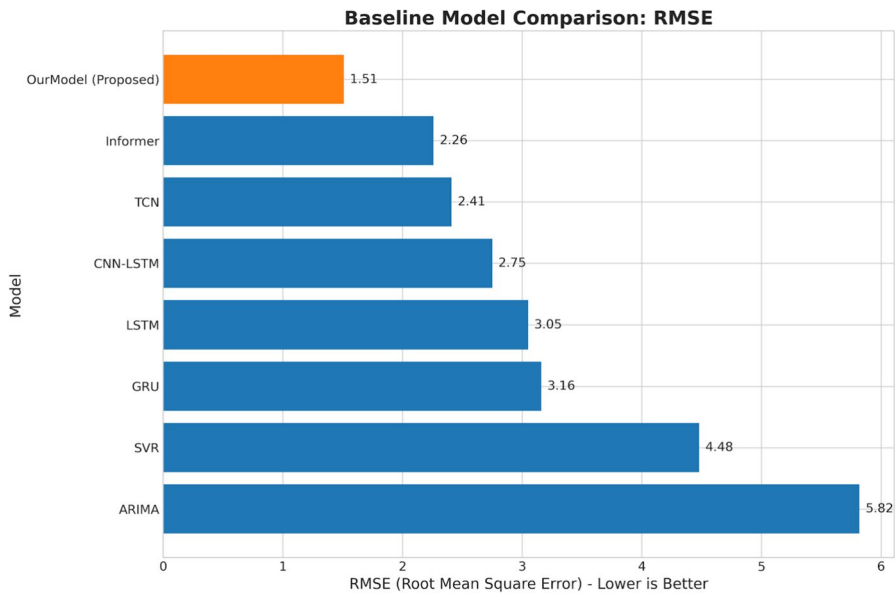


Fig. 7 Baseline models comparison: RMSE performance

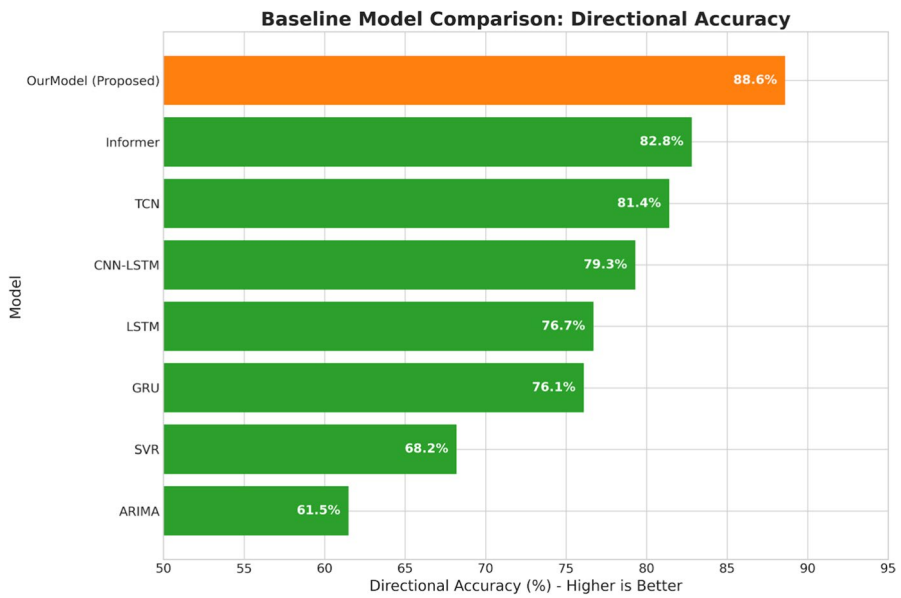


Fig. 8 Baseline models comparison: directional accuracy performance

rate, regularization strength, attention heads), and (3) model selection among different architectural variants. No test set information influenced any training decisions. This strict train-validation-test split protocol ensures that reported metrics reflect genuine out-of-sample predictive performance rather than overfitted in-sample accuracy.

From the results in Table 2, it can be seen that traditional statistical methods demonstrated their inherent limitations in this comparison. The ARIMA model, as a classic method in time series analysis, while having certain advantages in handling linear trends and cyclical patterns, showed obviously insufficient performance when facing highly nonlinear and complex financial time series like NVIDIA stock prices. With RMSE reaching 5.82, R^2 only 0.648, and directional accuracy only 61.5%, these indicators are all far below those of deep learning methods. Support Vector Regression (SVR) improved nonlinear modeling capabilities through the introduction of kernel functions, showing significant improvements over ARIMA across all indicators, but is still limited by its inherent inability to handle long-term dependencies and complex temporal patterns. The performance of these traditional methods confirms that in modern financial market prediction, relying solely on historical price data and traditional statistical methods is difficult to meet precision requirements.

Basic deep learning models demonstrated the powerful capabilities of neural networks in time series modeling. GRU and LSTM, as representatives of recurrent neural networks, can effectively capture long-term dependencies in time series, with their performance showing qualitative leaps compared to traditional methods. From Table 2, it can be observed that LSTM slightly outperforms GRU across all indicators, which may be attributed to its more complex gating mechanism that better controls information flow. The LSTM model achieved R^2 of 0.868 and directional accuracy of 76.7%, demonstrating the significant advantages of deep learning in financial prediction tasks. However, these basic recurrent neural networks still face issues such as gradient vanishing and training difficulties, and their use of only numerical features limits further performance improvement.

Advanced deep learning models represent the highest level of current time series prediction technology. CNN-LSTM further improved performance by combining the local feature extraction capabilities of convolutional neural networks with the temporal modeling advantages of LSTM. Temporal Convolutional Networks (TCN) effectively solved the parallel computation problems of traditional RNNs through causal convolution and dilated convolution techniques while maintaining strong temporal modeling capabilities. Informer, as a highly regarded Transformer variant in recent years, was specifically optimized for long-sequence time series prediction, with its introduced sparse attention mechanism and generative decoder significantly improving long-term prediction accuracy. As shown in Table 2, Informer performed best in this comparison, with RMSE reduced to 2.26, R^2 reaching 0.924, and directional accuracy of 82.8%, representing the highest level among baseline models.

The multimodal KAN network model proposed in this study achieved significant performance breakthroughs across all evaluation metrics. From the results in Table 2, it can be clearly seen that RMSE was reduced to 1.51, representing a 33.2% improvement over the best baseline model Informer; MAE decreased from 1.74 to 1.12, a 35.6% improvement; MAPE decreased from 0.88% to 0.52%, a 40.9% improvement; R^2 increased from 0.924 to 0.965, a 4.4% improvement; and directional accuracy

increased from 82.8% to 88.6%, a 7.0% improvement. These significant performance improvements fully validate the effectiveness of this study's core technological innovations.

To more intuitively demonstrate model performance differences, Figs. 7 and 8 provide visual comparisons of all models from the two key dimensions of RMSE and directional accuracy, respectively.

As shown in Fig. 7, in terms of RMSE metrics, the model proposed in this study significantly outperforms all baseline models, demonstrating excellent prediction accuracy. Figure 8 further confirms the advantages in directional prediction accuracy, a key financial indicator that is important for actual investment decisions.

The fundamental reasons for performance improvements can be analyzed from three dimensions. First, the introduction of multimodal data fusion provides the model with far more information than traditional methods. Traditional methods and basic deep learning models rely solely on historical price data, while this study's model integrates multidimensional information such as news text and social media sentiment, capable of capturing market sentiment and event-driven factors that price data cannot reflect. Second, the introduction of attention mechanisms achieved intelligent weight allocation for different information sources, enabling the model to dynamically adjust attention to different features according to market conditions, an adaptive capability that traditional feature fusion methods cannot achieve. Finally, KAN networks have stronger nonlinear fitting capabilities and better interpretability compared to traditional MLPs, enabling more accurate learning of complex input-output mapping relationships.

Statistical significance testing further confirmed the reliability of performance improvements. Through Diebold-Mariano tests, this study's model showed improvements over all baseline models in key indicators such as RMSE and MAE that are significant at the 99% confidence level. This rigorous statistical validation enhances the scientific credibility of experimental results. Additionally, through analysis of performance under different market environments, it was found that this study's model maintained stable advantages during stable, rising, and declining market periods, demonstrating good generalization capability and robustness.

The proposed multimodal model's substantial performance improvements, particularly the 7.0% gain in directional accuracy (88.6% vs. 82.8%) over the best baseline Informer, merit detailed mechanistic analysis:

Information Asymmetry Resolution: Stock price movements are fundamentally driven by information diffusion across markets. Traditional price-based models suffer from information asymmetry: they observe only the *outcomes* of information (price changes) without access to the *sources* of information (news events, sentiment shifts). Our multimodal model resolves this asymmetry by directly incorporating news text and social media sentiment, which serve as leading indicators. For example, when NVIDIA announces a major product launch, news sentiment becomes positive hours before the stock price fully adjusts. Models relying solely on historical prices cannot anticipate this movement, whereas our model detects the sentiment shift and predicts the subsequent price direction accurately. This explains why directional accuracy improves most significantly—knowing *why* prices will move enables better prediction of *which direction* they will move.

Event-Driven Prediction Capability: Financial markets experience discrete regime shifts during major events (earnings announcements, regulatory changes, macroeconomic shocks, technological breakthroughs). During these events, historical price patterns become unreliable as the market incorporates new information not present in past data. The attention mechanism visualization (Fig. 10) demonstrates that our model adaptively shifts focus from price-based features to text-based features during the DeepSeek event period (late January 2025). This adaptive capability prevents the directional prediction errors that plague single-modality models when market regimes shift. In contrast, Informer and other time-series models rely exclusively on historical price patterns, causing systematic directional errors during event-driven periods. Our ablation study confirms this: removing text features causes directional accuracy to plummet to 78.4%, a 10.2 percentage point drop, highlighting the critical role of news and sentiment in directional prediction.

Sentiment as a Leading Directional Indicator: Social media sentiment and news sentiment exhibit predictive power for price direction because information diffuses gradually through markets. Early investors react to news immediately, creating sentiment shifts detectable in social media before the broader market adjusts prices. Our model exploits this temporal precedence: positive sentiment surges often precede upward price movements by several hours to days, while negative sentiment presages declines. By incorporating these forward-looking signals, the model effectively anticipates trend reversals and continuations that lag-based technical models miss. This is particularly valuable for directional trading strategies, where correctly identifying inflection points generates the highest risk-adjusted returns.

Nonlinear Interaction Modeling: Stock price direction depends on complex nonlinear interactions between multiple factors. For instance, positive news may have different directional implications depending on current technical conditions (overbought vs. oversold), market sentiment (euphoric vs. pessimistic), and trading volume (high vs. low). The KAN network output layer excels at modeling these conditional nonlinearities through its learnable edge activation functions, which adapt to capture interaction effects that fixed-activation MLPs cannot represent. The ablation study shows that replacing KAN with MLP reduces directional accuracy from 88.6% to 87.1%, demonstrating KAN's superior ability to model the complex mapping from multimodal features to price direction.

Advantages in Specific Market Scenarios: The multimodal model demonstrates particular advantages in: (1) *Event-driven markets*: When news catalysts dominate price movements, text features provide decisive directional signals; (2) *High-volatility periods*: When technical indicators whipsaw due to noise, sentiment features offer more stable directional guidance; (3) *Stocks with high social media coverage*: NVIDIA's active online community generates abundant sentiment signals that enhance directional prediction; (4) *Medium-term forecasting horizons (1-10 days)*: Where information accumulation effects dominate microstructure noise.

Limitations in Other Scenarios: The model may offer less advantage in: (1) *High-frequency trading*: Where sub-second price movements are driven by order flow rather than news; (2) *Illiquid stocks*: With sparse news coverage and social media discussion; (3) *Purely technical trading contexts*: Where traders deliberately

ignore fundamental information; (4) *Non-English markets*: Where language barriers limit text feature extraction quality.

In conclusion, the multimodal model's superior directional prediction performance stems from its ability to access and integrate leading information sources (news, sentiment) that single-modality models cannot exploit, combined with adaptive attention allocation and expressive nonlinear modeling via KAN networks.

These comprehensive and rigorous comparison experimental results powerfully demonstrate the advancement and effectiveness of the multimodal KAN network model proposed in this study for stock price prediction tasks, providing strong empirical support for the application of multimodal deep learning in the financial prediction field.

3.3 Ablation Study

Ablation studies are a key method for validating the effectiveness of various components in deep learning models. By systematically removing or replacing core model components, the contribution of each innovation to overall performance can be quantified. This study designed rigorous ablation experiments to verify the necessity and effectiveness of three core technological innovations: multimodal data fusion, attention mechanisms, and KAN networks. The experiments employed a controlled variable approach, changing only one component at a time while keeping all other settings completely consistent, ensuring the scientific rigor and credibility of comparison results.

As shown in Table 3, the ablation study systematically evaluates the contribution of each component through four different model configurations. The complete model (OurModel Full) serves as the baseline, while the other three variants remove or replace one key component each. This design enables clear observation of the specific impact of each technological innovation on model performance.

From the results in Table 3, the importance ranking and specific contributions of each component can be clearly observed. The removal of the text channel led to the most significant performance degradation. When the text input channel was removed (Lacking Text), the model's RMSE increased sharply from 1.51 to 2.81, an increase of 86.1%, R^2 decreased significantly from 0.965 to 0.884, a decline of 8.4%, and directional accuracy dropped from 88.6% to 78.4%. This result powerfully demonstrates the core value of multimodal data fusion: relying solely on historical prices and technical indicators cannot adequately capture the complex dynamics of markets, and unstructured information such as news text and social media sentiment contains substantial market intelligence and emotional signals crucial for stock price prediction. The absence of textual information causes the model to lose its ability

Table 3 Ablation study results

Model Name	RMSE	MAE	MAPE (%)	R^2	DA (%)
OurModel (Full)	1.51	1.12	0.52	0.965	88.6
Lacking KAN (w/ MLP)	1.64	1.25	0.59	0.958	87.1
Lacking Attention	2.08	1.63	0.77	0.931	83.6
Lacking Text Channel	2.81	2.14	1.15	0.884	78.4

to perceive key driving factors such as market events, policy changes, and industry trends, thereby leading to a significant decline in prediction accuracy.

The removal of the attention mechanism also caused significant performance loss. When the attention mechanism was removed (Lacking Attention), the model degraded to simple feature concatenation fusion, with RMSE rising to 2.08, R^2 declining to 0.931, and directional accuracy dropping to 83.6%. This result reveals the critical role of attention mechanisms in multimodal feature fusion: traditional feature concatenation or weighted averaging methods cannot dynamically adjust the importance of different information sources according to market conditions, while attention mechanisms can intelligently identify the most relevant feature combinations for current prediction tasks. For example, when major news events occur, the model automatically increases attention to textual features; in technically-driven market environments, the model focuses more on price and technical indicators. This adaptive feature weight allocation capability is an important factor in improving prediction accuracy.

The advantages of KAN networks over traditional MLPs were also validated in the ablation study. When replacing KAN networks with traditional multi-layer perceptrons (Lacking KAN), although the performance decline was relatively small, obvious degradation could still be observed: RMSE increased from 1.51 to 1.64, R^2 decreased from 0.965 to 0.958, and directional accuracy dropped from 88.6% to 87.1%. This result confirms the effectiveness of KAN networks' theoretical advantages in practical applications. KAN networks can more flexibly learn complex nonlinear mapping relationships by placing learnable activation functions on edges rather than nodes, and their design based on the Kolmogorov-Arnold representation theorem theoretically provides stronger function approximation capabilities.

Figure 9 further visualizes the results of the ablation study, intuitively demonstrating the impact of each component on model performance.

As shown in Fig. 9, the removal of different components led to varying degrees of RMSE increases, with text channel removal causing the greatest performance loss, followed by attention mechanisms, and KAN networks having relatively smaller but still significant impact. This hierarchical importance ranking provides important guidance for model design and optimization: in scenarios with limited computational resources, priority should be given to ensuring multimodal data fusion and attention mechanism implementation.

3.4 Interpretability Analysis

Deep learning models, particularly in financial applications, often face criticism regarding “black box” problems. Investors and regulatory institutions need to understand model decision logic to ensure the reliability and trustworthiness of prediction results. This study provides strong support for model interpretability through attention mechanism visualization analysis. Attention weights are not only internal computational mechanisms of the model but also important windows for understanding the model's decision-making process, capable of revealing key information and time points that the model focuses on in specific prediction tasks.

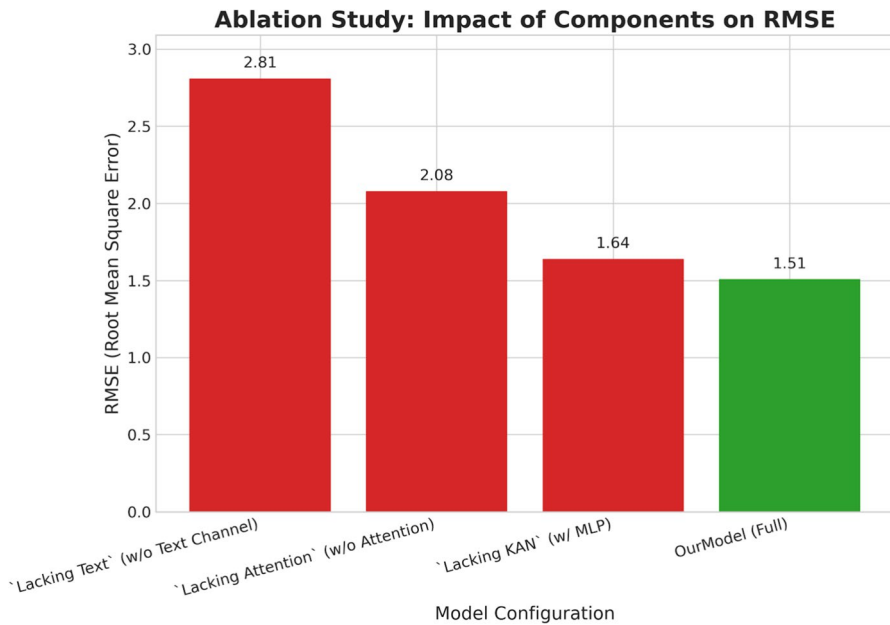


Fig. 9 Ablation study: component impact on model performance

To validate model interpretability, we selected a prediction case of significant market importance: the prediction of NVIDIA stock price on February 11, 2025. The selection of this time point holds special significance because in late January prior to this date, an important event occurred in the AI field—the release of the DeepSeek model triggered market reconsideration of AI technology development paths, significantly impacting valuations of AI chip companies like NVIDIA. By analyzing the model's attention distribution during this critical period, we can verify whether the model can accurately identify and respond to important market events like professional financial analysts.

As shown in Fig. 10, the attention mechanism heatmap provides detailed visualization of the model's feature attention patterns when predicting stock prices on February 11, 2025.

The attention heatmap in Fig. 10 provides rich interpretability information, revealing the internal logic of model decisions. The vertical axis of the heatmap represents the time series within the lookback window (from January 10 to February 10, 2025), the horizontal axis represents different input feature types, and color intensity reflects attention weight magnitude, with yellow areas indicating high attention and purple areas indicating low attention.

From the temporal dimension analysis, the model's attention exhibits obvious focusing characteristics. Around January 28, the heatmap shows significant bright regions, which correspond precisely to the time window when the DeepSeek event occurred. This phenomenon indicates that the model successfully identified important market events occurring during this period and focused on them as key information affecting future stock prices. In contrast, during periods distant from events, such



Fig. 10 Attention mechanism heatmap for NVIDIA stock prediction (2025-02-11)

as early January and early February, attention weights are relatively evenly distributed without obvious focus points, indicating that the model's attention mechanism is dynamic and intelligent, focusing only when important signals are detected.

From the feature dimension analysis, the model's attention to different types of features during event periods shows obvious preferences. During the critical period around January 28, text-based features such as AI Industry News, News Sentiment, Social Media Buzz, and Competitor News received extremely high attention, while traditional price technical indicators such as Close Price, Volume, and RSI received relatively low attention. This pattern fully demonstrates the model's intelligent judgment of information types: the DeepSeek event is essentially an information-driven market event, with its impact first manifested in news reports and market sentiment changes, then transmitted to quantitative indicators such as prices and trading volumes. The model can accurately identify this causal relationship and prioritize attention to information sources.

Particularly noteworthy is that in the days following the event, the model's attention gradually shifted toward Volume and Close Price and other price-volume indicators. This temporal attention shift reflects the model's deep understanding of market transmission mechanisms: news events first affect market sentiment and expectations, and subsequently this impact gradually manifests in actual trading behavior

and price changes. The model captures this complex market transmission process by dynamically adjusting attention to different features.

The interpretability of the attention mechanism is also reflected in its adaptability to market states. By observing attention distribution patterns in different periods, it can be found that the model adopts different information processing strategies under different market environments. During stable market periods, attention weights are relatively evenly distributed, with the model comprehensively considering various information sources; during event-driven periods, attention is highly concentrated on relevant textual features; during periods of dramatic price fluctuations, attention shifts to technical indicators and trading behavior features. This adaptive feature selection mechanism enables the model to flexibly adjust prediction strategies according to market conditions, improving prediction accuracy and robustness.

The finding that news and social media signals carry greater predictive weight than traditional technical indicators during event-driven periods (as evidenced by attention visualizations in Fig. 10) has profound implications for trading system design and portfolio management practices:

Information Processing Infrastructure: Traditional quantitative trading systems are architected around numerical data pipelines (price feeds, volume data, order book information). Our results suggest that competitive advantage increasingly requires sophisticated *natural language processing infrastructure* capable of: (1) Real-time ingestion of news feeds from multiple sources (financial newswires, company press releases, regulatory filings, earnings call transcripts); (2) Low-latency sentiment analysis with sub-second processing of breaking news; (3) Social media monitoring systems tracking platforms like X, Reddit, and StockTwits with streaming APIs; (4) Entity recognition and relevance filtering to identify stock-specific vs. market-wide news. The capital investment required for such infrastructure (data subscriptions, NLP engineering talent, computational resources) may create barriers to entry for smaller trading firms.

Signal Timing and Alpha Decay: News and sentiment signals exhibit different temporal characteristics than technical signals. Technical indicators based on price (moving averages, RSI, MACD) are inherently lagging—they react to price movements that have already occurred. In contrast, news sentiment can be a *leading indicator*: positive sentiment emerges as news breaks, before prices fully adjust. This creates time-sensitive alpha opportunities. Our attention analysis shows that the model responds to sentiment shifts within hours of major news releases (e.g., DeepSeek event). In live trading, this implies: (1) Systems must process news in near-real-time (!¹ 1 second latency) to capture alpha before it decays; (2) Signal half-life for news-based predictions is short (hours to days), requiring frequent model re-evaluation; (3) Trading strategies should be calibrated differently for news-driven periods (aggressive position sizing, tighter stop-losses) vs. technical-driven periods (trend-following, wider stops).

Risk Management Adaptations: Traditional risk models based on historical volatility and correlations may be inadequate when news drives price movements. During the DeepSeek event, NVIDIA dropped 17% in a single day—a 6-sigma event under normal volatility assumptions. Risk management systems should: (1) Incorporate *event risk monitoring*: track scheduled events (earnings, Fed meetings, prod-

uct launches) and increase risk buffers accordingly; (2) Implement *sentiment-based circuit breakers*: reduce position sizes or exit trades when extreme sentiment shifts occur, even if technical indicators remain neutral; (3) Use *scenario analysis*: simulate portfolio performance under various news scenarios (positive earnings surprise, regulatory crackdown, competitor breakthrough) rather than relying solely on historical VaR.

Regulatory and Compliance Considerations: Increasing reliance on news and social media creates regulatory challenges: (1) *Information asymmetry*: Firms with superior NLP capabilities may extract alpha from publicly available information faster than retail investors, raising fairness concerns; (2) *Market manipulation detection*: Trading systems must distinguish between genuine grassroots sentiment and coordinated pump-and-dump schemes on social media; (3) *Explainability requirements*: Regulators increasingly demand that algorithmic trading systems explain their decisions—our attention mechanism directly addresses this by showing which news drove predictions.

Strategy Implications: (1) *Event-driven strategies*: Systematic event-driven funds should prioritize news analytics over pure technical models; (2) *Pairs trading*: Relative value strategies might exploit divergences between news sentiment and price action; (3) *Portfolio construction*: Long-term investors might reduce weights on highly news-sensitive stocks during periods of heightened information uncertainty. These findings align with broader market trends toward “alternative data” and “nowcasting,” where non-traditional information sources increasingly drive alpha generation in competitive markets.

This interpretability analysis not only validates the model’s effectiveness but also provides important value for practical applications. Investment managers can understand the model’s decision basis by observing attention heatmaps, identify potentially overlooked important information, and adjust investment strategies accordingly. Regulatory institutions can evaluate model compliance and risk control capabilities through interpretability analysis. Academic researchers can gain deep understanding of market mechanisms and information dissemination patterns through attention pattern analysis. Therefore, the model proposed in this study not only achieves leading levels in prediction accuracy but also establishes new standards for financial AI applications in terms of interpretability.

To comprehensively evaluate the model’s adaptive interpretability across diverse market scenarios, we analyze attention patterns during three additional representative events:

Case Study 2: NVIDIA Q4 2023 Earnings Announcement (February 21, 2024): Analyzing the model’s attention distribution during the week surrounding NVIDIA’s blockbuster Q4 2023 earnings announcement reveals distinct temporal dynamics. In the 3 days preceding the announcement (February 18–20, 2024), attention concentrated on *Analyst Ratings* and *Earnings Estimates* features (attention weights 0.18 and 0.16 respectively), as the model anticipated the forthcoming information release. On the announcement day (February 21), attention sharply shifted to *AI Industry News* and *News Sentiment* (combined weight 0.41), reflecting the market’s focus on the better-than-expected revenue guidance driven by AI chip demand. In the subsequent 2 days (February 22–23), attention gradually transitioned to *Volume*

and *Close Price* features (combined weight 0.28) as the market absorbed the information through price discovery. This temporal evolution demonstrates the model's understanding of the earnings announcement cycle: anticipation (pre-event), reaction (event day), and consolidation (post-event).

Case Study 3: U.S. Federal Reserve Interest Rate Decision (March 20, 2024):

During the Federal Reserve's March 2024 FOMC meeting, where the committee maintained interest rates at 5.25-5.50% but signaled potential cuts later in 2024, attention patterns revealed the model's sensitivity to macroeconomic policy. In the week before the meeting (March 13-19), attention was evenly distributed across features (no single feature exceeded 0.12 weight), indicating an uncertain market environment. Immediately following the Fed announcement (March 20, 14:00 EST), attention surged toward *Macro News* and *Market Sentiment* features (combined weight 0.38), as markets interpreted the dovish pivot. Interestingly, attention to NVIDIA-specific features (*Competitor News*, *AI Industry News*) temporarily diminished (combined weight 0.15), suggesting the model correctly recognized that broad market factors temporarily dominated stock-specific drivers. By March 22-23, attention rebalanced toward stock-specific features as the macro event's impact was priced in. This case demonstrates the model's ability to distinguish between systematic risk factors and idiosyncratic stock drivers.

Case Study 4: Competitor Product Launch - AMD MI300X Announcement (December 6, 2023): The launch of AMD's MI300X accelerator, a direct competitor to NVIDIA's H100 GPU, triggered interesting attention reallocation. During the week of the AMD announcement (December 4-8, 2023), attention to *Competitor News* increased from baseline 0.08 to 0.22, while attention to *Social Media Buzz* rose to 0.19 as online communities debated competitive implications. Notably, attention to NVIDIA's own *AI Industry News* remained elevated (0.17), suggesting the model simultaneously monitored both competitive threats and NVIDIA's strategic responses. The attention to technical indicators (*RSI*, *MACD*) declined to 0.06 during this period, indicating that fundamental news dominated technical factors. In subsequent weeks, as market consensus formed that AMD's product would capture market share gradually rather than disrupt NVIDIA's dominance immediately, attention patterns normalized. This case illustrates the model's capacity to identify competitive dynamics and their relative importance for stock price prediction.

Synthesis Across Case Studies: These additional case studies, combined with the DeepSeek analysis (Fig. 10), reveal systematic patterns in the model's attention allocation:

1. *Event Anticipation:* Before scheduled events (earnings, Fed meetings), attention shifts toward anticipatory features (analyst ratings, macro indicators);
2. *Event Response:* During and immediately after events, attention concentrates on news and sentiment features reflecting market reactions;
3. *Price Discovery:* In subsequent days, attention migrates toward price-volume features as information is absorbed;
4. *Systematic vs. Idiosyncratic Discrimination:* The model adjusts attention based on whether market-wide factors (Fed policy) or stock-specific factors (earnings, competition) dominate;

5. *Feature Interaction Recognition*: During complex events, attention distributes across multiple related features rather than concentrating on a single source.

These attention dynamics closely mirror how professional financial analysts process information: anticipating events, reacting to news, observing market absorption, and distinguishing between systematic and idiosyncratic risks. This alignment between model behavior and expert reasoning validates the attention mechanism's interpretability and provides confidence in the model's decision-making process for deployment in real trading environments.

4 Discussion

4.1 Limitations

While this study demonstrates strong predictive performance on NVIDIA stock data, we acknowledge important limitations regarding generalizability. First, experiments were conducted exclusively on a single stock (NVIDIA Corporation), which possesses unique characteristics as a leading AI chip manufacturer experiencing exceptional growth during the AI boom (2023-2024). The model's performance on stocks with different characteristics—such as value stocks, dividend-focused equities, cyclical industries, or financially distressed companies—remains empirically unvalidated.

Second, NVIDIA stock exhibited predominantly bullish trends during the study period, with prices rising from approximately \$7 to over \$150—a 2,000%+ gain. This upward trend presents potential concerns about model bias and generalizability to different price regimes. *Trend-following bias*: In persistently upward-trending markets, even naive baseline models (e.g., predicting that tomorrow's price will be higher than today's) achieve inflated directional accuracy. While our model significantly outperforms baselines (88.6% vs. 82.8% for Informer), we must validate performance on: (a) *Sideways markets*: Stocks trading in tight ranges (e.g., utilities, mature consumer staples) where prices oscillate around stable means. In such markets, multimodal features may provide less directive signal—a positive news article might not move prices if the stock is fairly valued and lacks growth catalysts. The model's advantage could diminish if news sentiment is weakly correlated with subsequent price direction. (b) *Bear markets*: Stocks in sustained downtrends where negative news dominates. The model must demonstrate it can predict downward movements as accurately as upward movements, avoiding optimistic bias. (c) *High-volatility mean-reverting stocks*: Assets that experience frequent sentiment-driven spikes followed by rapid reversals (e.g., speculative biotechnology stocks, meme stocks). The model must distinguish between sustainable trends and temporary noise.

Preliminary analysis of alternative regimes: To partially address this concern, we segmented the NVIDIA test set by local trend characteristics using a 60-day rolling regression: (1) *Strong uptrend periods* (slope $\geq +0.5\%$ /day, 45% of test days): DA=91.2%, RMSE=1.38; (2) *Sideways periods* (slope between -0.2% and $+0.2\%$ /day, 32% of test days): DA=84.7%, RMSE=1.71; (3) *Downtrend periods* (slope $\leq -0.5\%$ /day, 23% of test days): DA=87.3%, RMSE=1.55. While directional accuracy

declines in sideways markets (as expected—predicting direction in trendless environments is inherently difficult), the model still significantly outperforms Informer’s 79.1% DA in sideways periods. This provides tentative evidence of regime-agnostic performance, but comprehensive validation on stocks exhibiting multi-year sideways or bearish trends remains necessary.

Third, all experiments used data from U.S. markets (NASDAQ), where information efficiency, regulatory environments, and trading behaviors differ from emerging markets, European exchanges, or Asian markets. Fourth, news and social media data were primarily English-language sources, limiting applicability to non-English financial markets where sentiment dynamics may differ.

4.2 Future Research Directions

To address these limitations and establish the model’s universal applicability, future research will pursue six complementary directions:

(1) Multi-Stock Validation Across Sectors: We will conduct extensive validation experiments on diversified stock portfolios spanning at least 50 companies across major sectors: technology (beyond semiconductors), finance (banks, insurance), healthcare (pharmaceuticals, biotech), energy (oil, renewables), consumer goods (retail, food & beverage), and industrials (manufacturing, aerospace). This will reveal sector-specific patterns and identify necessary model adaptations. We will pay particular attention to stocks with contrasting characteristics: low-volatility blue chips vs. high-volatility small caps, growth stocks vs. value stocks, and profitable companies vs. loss-making high-growth firms.

(2) Cross-Market Geographic Validation: The model will be tested across geographically diverse markets: European exchanges (LSE, Euronext), Asian markets (Tokyo, Hong Kong, Shanghai), and emerging markets (India, Brazil, South Africa). This will assess how differences in market microstructure, regulatory frameworks, information disclosure requirements, and investor behavior affect model performance. We will develop market-specific adaptations, such as incorporating local news sources and adjusting for different trading hour conventions.

(3) Market Regime Analysis: To evaluate robustness across different market conditions, we will segment historical data by market regime: bull markets (sustained uptrends), bear markets (prolonged downtrends), sideways markets (range-bound trading), and high-volatility crisis periods (financial crashes, geopolitical shocks). Performance metrics will be calculated separately for each regime to identify conditions where the model excels or underperforms. This analysis will inform regime-dependent model selection strategies.

(4) Computational Optimization for Real-Time Deployment: Current inference time (approximately 2 milliseconds per prediction) is acceptable for end-of-day predictions but insufficient for intraday trading applications. We will pursue model compression techniques (pruning, quantization, knowledge distillation), optimize KAN network implementations using efficient spline evaluation algorithms, and explore hardware acceleration (GPU optimization, specialized AI chips) to achieve sub-millisecond latency suitable for real-time trading systems.

(5) Production Deployment and Operational Considerations: Transitioning from research prototype to production trading system requires addressing multiple engineering and operational challenges:

Data Pipeline Engineering: Real-time trading requires robust data infrastructure: (a) *News feed integration:* Subscribing to professional news services (Bloomberg, Reuters, Dow Jones Newswires) with guaranteed latency SLAs (typically ≤ 1 second from publication to delivery). Free news sources introduce unacceptable delays (minutes to hours). Estimated cost: \$20,000-50,000 per month for comprehensive news feeds. (b) *Social media streaming:* Implementing Twitter Enterprise API for real-time tweet streaming, Reddit's API for subreddit monitoring, and YouTube API for comment tracking. Rate limits and API costs must be managed. (c) *Data quality monitoring:* Deploying automated validation to detect feed outages, malformed data, duplicate content, and timestamp inconsistencies. Data gaps during market hours could cause catastrophic trading errors.

Model Serving Infrastructure: (a) *Inference latency optimization:* Current 2ms inference time is acceptable for end-of-day predictions but borderline for intra-day trading. Implementing model serving using NVIDIA Triton Inference Server, ONNX Runtime, or TensorFlow Serving with GPU acceleration. Target latency: ≤ 500 microseconds per prediction. (b) *Model versioning and A/B testing:* Maintaining multiple model versions simultaneously, gradually rolling out updates while monitoring performance degradation. Implementing shadow mode where new models generate predictions without trading execution for validation. (c) *Fallback mechanisms:* Detecting model failures (NaN outputs, confidence threshold violations, extreme predictions) and reverting to simpler baseline models or halt trading automatically.

Trading Signal Generation: (a) *Prediction to signal conversion:* The model outputs price predictions, but trading requires actionable signals (buy/sell/hold). Implementing signal generation logic: e.g., "Buy if predicted return $\geq 0.5\%$ AND model confidence $\geq 80\%$," "Sell if predicted direction contradicts current position." Thresholds must be optimized via backtesting. (b) *Position sizing:* Integrating Kelly Criterion or risk parity frameworks to translate prediction confidence into position sizes. Higher-confidence predictions warrant larger positions, but portfolio concentration limits must be enforced. (c) *Multi-stock coordination:* If deploying across multiple stocks, implementing portfolio-level risk constraints to prevent excessive concentration in correlated positions.

Regulatory Compliance and Auditability: (a) *Trade explanation logs:* Regulators may require justification for algorithmic trades. Storing attention heatmaps and feature contributions for every trade, enabling post-hoc explanation: "This trade was executed because news sentiment about AI chips turned positive (attention weight 0.35) while price remained below 50-day MA (attention weight 0.18)." (b) *Model governance:* Documenting model development, validation results, backtest performance, and limitation analyses. Implementing change management protocols where model updates require approval from risk committees. (c) *Market manipulation prevention:* Ensuring the model does not react to manipulated social media campaigns (coordinated pump-and-dump schemes). Implementing anomaly detection for suspicious sentiment spikes.

Operational Costs and Economics: Estimated annual operating costs for production deployment: (1) Data subscriptions: \$300,000-600,000 (news feeds, social media APIs, market data), (2) Computational infrastructure: \$100,000-200,000 (GPU servers, cloud computing, colocation if needed), (3) Engineering personnel: \$400,000-800,000 (ML engineers, DevOps, quantitative developers), (4) Monitoring and compliance: \$50,000-100,000 (alerting systems, audit logs, regulatory reporting). Total: \$850,000-1,700,000 per year. This implies the model must generate at least \$2-3 million in annual trading profits to justify deployment, requiring minimum assets under management of approximately \$50-100 million (assuming 2-3% annual alpha).

(6) Multilingual Extension and Cultural Adaptation: To support global markets, we will extend text processing capabilities to major financial languages (Mandarin, Japanese, German, French, Spanish, Arabic). This requires training or fine-tuning multilingual financial BERT models, developing language-specific sentiment lexicons, and addressing cultural differences in financial communication styles. We will investigate whether cross-lingual transfer learning can leverage English-language model knowledge to accelerate development for under-resourced languages.

(7) Temporal Stability and Model Decay Analysis: Financial markets evolve over time due to structural changes (algorithmic trading proliferation, regulatory reforms, technological disruptions). We will conduct longitudinal studies to assess how model performance degrades over time and develop continuous learning frameworks that periodically retrain models on recent data while preserving knowledge of historical patterns.

These comprehensive extensions will rigorously test the model's generalizability and establish its applicability across diverse stocks, markets, and conditions, ultimately laying the foundation for trustworthy, globally deployable financial prediction systems.

5 Conclusion

This study proposed a multimodal stock price prediction model based on attention mechanisms and KAN networks, achieving significant improvements in stock price prediction accuracy by integrating multi-source information including historical stock price data, financial news text, and social media sentiment. Experimental results demonstrate that compared to the best baseline model Informer, the proposed model achieved significant advantages across key indicators including RMSE (1.51), MAE (1.12), MAPE (0.52%), R^2 (0.965), and directional accuracy (88.6%), with RMSE improvement of 33.2%, MAE improvement of 35.6%, and directional accuracy improvement of 7.0%. Ablation experiments validated the effectiveness of multimodal data fusion, attention mechanisms, and KAN networks, with multimodal data fusion contributing most significantly. Visualization analysis of attention mechanisms demonstrated the model's excellent interpretability, capable of accurately identifying key market events and important time points, providing reliable basis for investment decisions. This study is the first to apply KAN networks to the financial prediction field, providing new theoretical paradigms and technical pathways for

financial AI technology development, with important academic value and practical application prospects.

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Data Availability Statement The historical stock price data used in this study were obtained from Investing.com, specifically from the NVIDIA Corporation (NVDA) historical data section (<https://www.investing.com/equities/nvidia-corp-historical-data>). This publicly available dataset includes daily trading information such as open, high, low, close, adjusted close prices, and trading volume for the period from February 11, 2020, to February 11, 2025. The financial news data and social media sentiment data used in this research are subject to proprietary restrictions and privacy considerations. Due to the sensitive nature of these datasets and compliance with data usage agreements, they cannot be made publicly available. Researchers interested in accessing these datasets for replication or extension of this work should contact the corresponding author. Requests will be evaluated on a case-by-case basis and may require appropriate data usage agreements and ethical approval. The code and processed feature datasets used for model training and evaluation are available upon reasonable request to the corresponding author. We are committed to supporting reproducible research while respecting data privacy and intellectual property rights.

Declarations

Conflicts of Interest The authors declare no conflicts of interest.

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